CS 6041
Theory of Computation

Pushdown Automata

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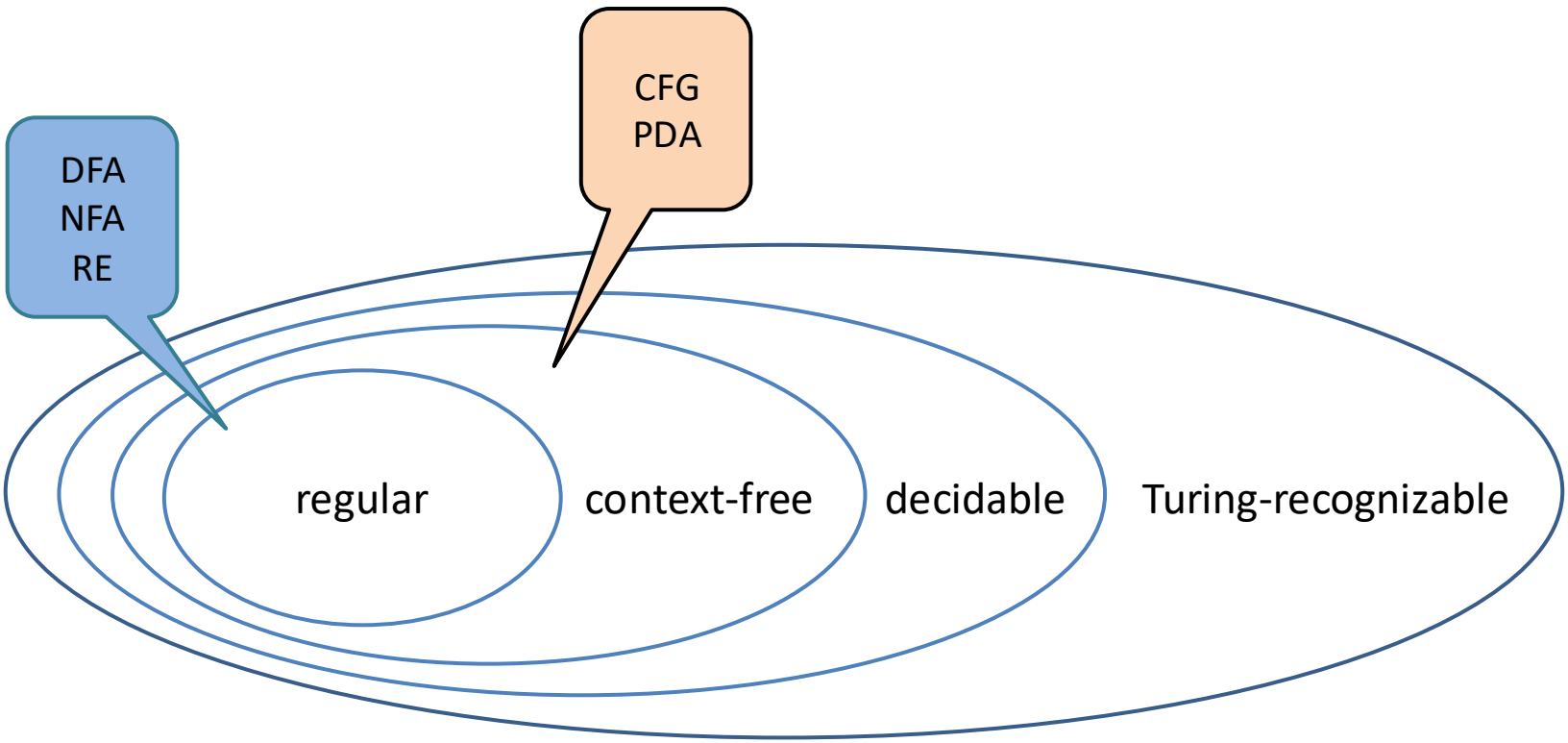
https://kevinsuo.github.io/

Pushdown Automata (PDA)



Equivalence of PDA and CFG

- Theorem: A language is context free **if and only if** some pushdown automaton recognizes it

- **A language is CFL $\Longrightarrow$ some PDA recognizes it**

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- Grammar G_1:

  $A \rightarrow 0A1$

  $A \rightarrow B$

  $B \rightarrow \#$

- Derivation: $A \Rightarrow 0A1 \Rightarrow$

  $00A11 \Rightarrow 000A111 \Rightarrow$

  $000B111 \Rightarrow 000\#111$

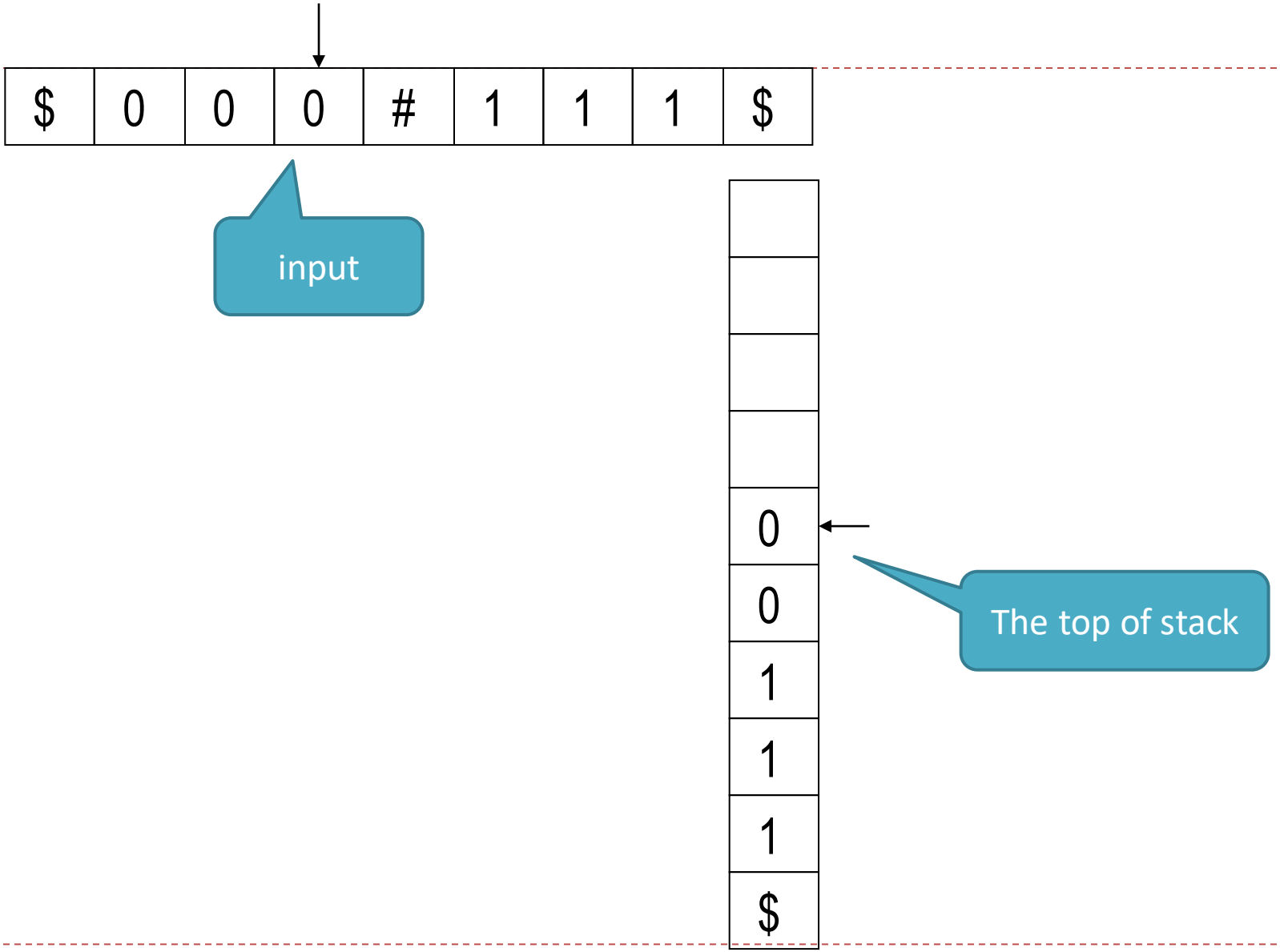
If a PDA can recognize strings in one CFL, done!

How to create a PDA to recognize 000#111

A language is CFL $\Longrightarrow$ some PDA recognizes it

- Details: how PDA recognizes a string of a CFL
  - **Step 1: Compare the input with the top of stack**
  - Step 2: If the top of stack is variable, then simulate the derivation
  - Step 3: If the top of stack is terminal, then do the match

Step 1: Compare the input with the top of stack



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  - Step 1: Compare the input with the top of stack
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Step 2: If the top of stack is variable, then simulate the derivation

$	0	0	0	#	1	1	1	$

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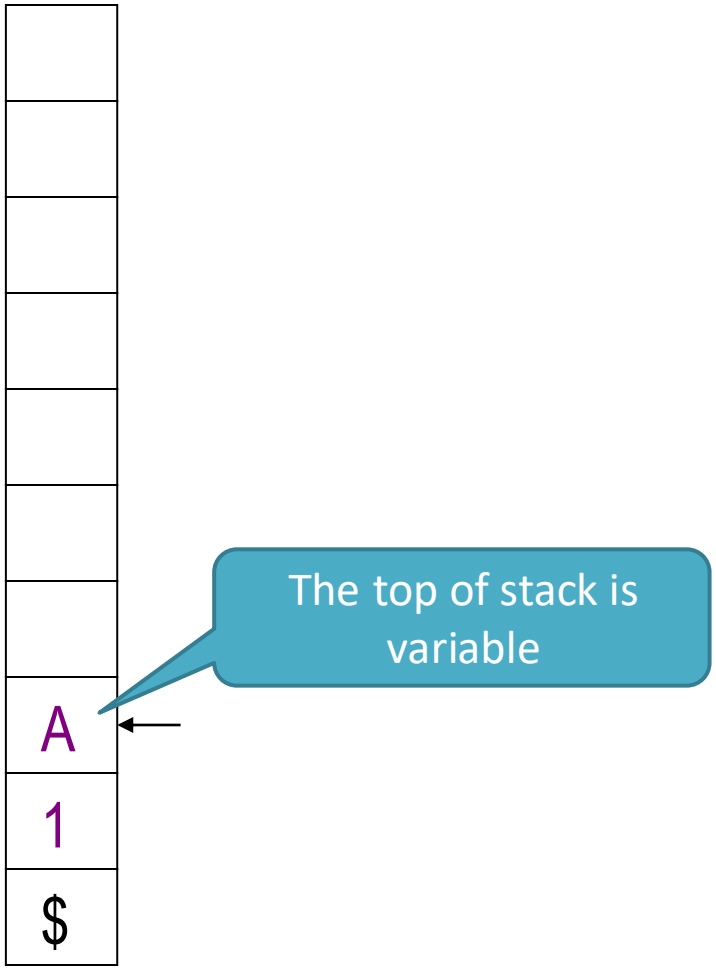
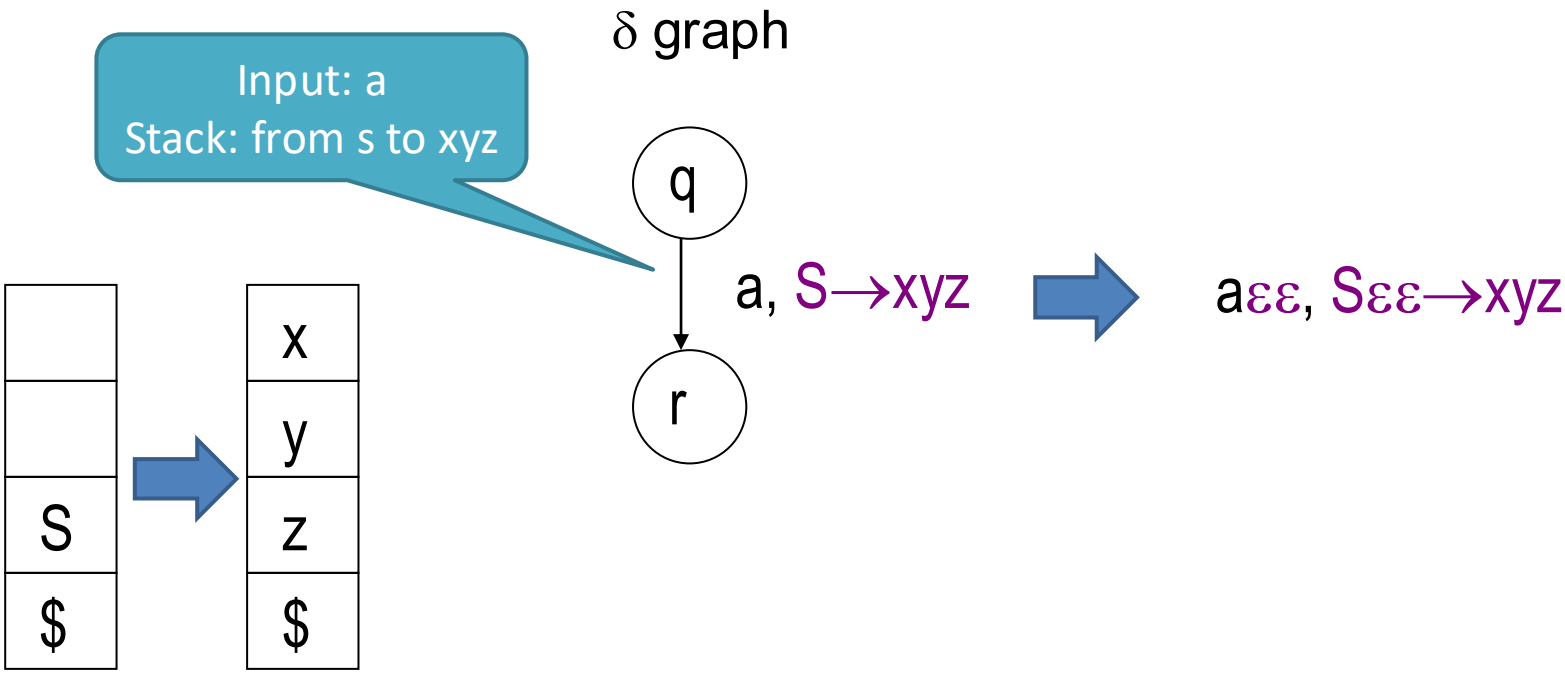


Step 2: If the top of stack is variable, then simulate the derivation

- Simulate the derivation $S \Rightarrow xyz$ based on rule $S \rightarrow xyz$



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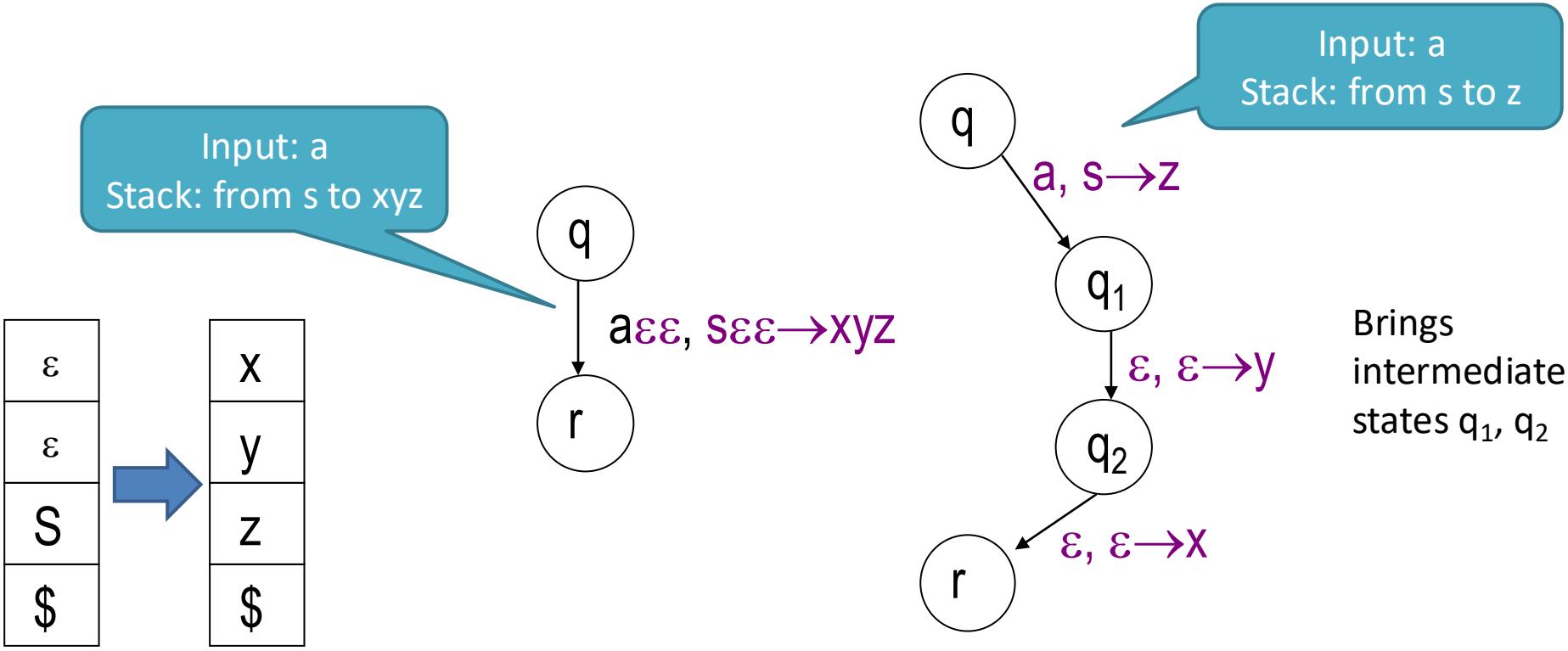
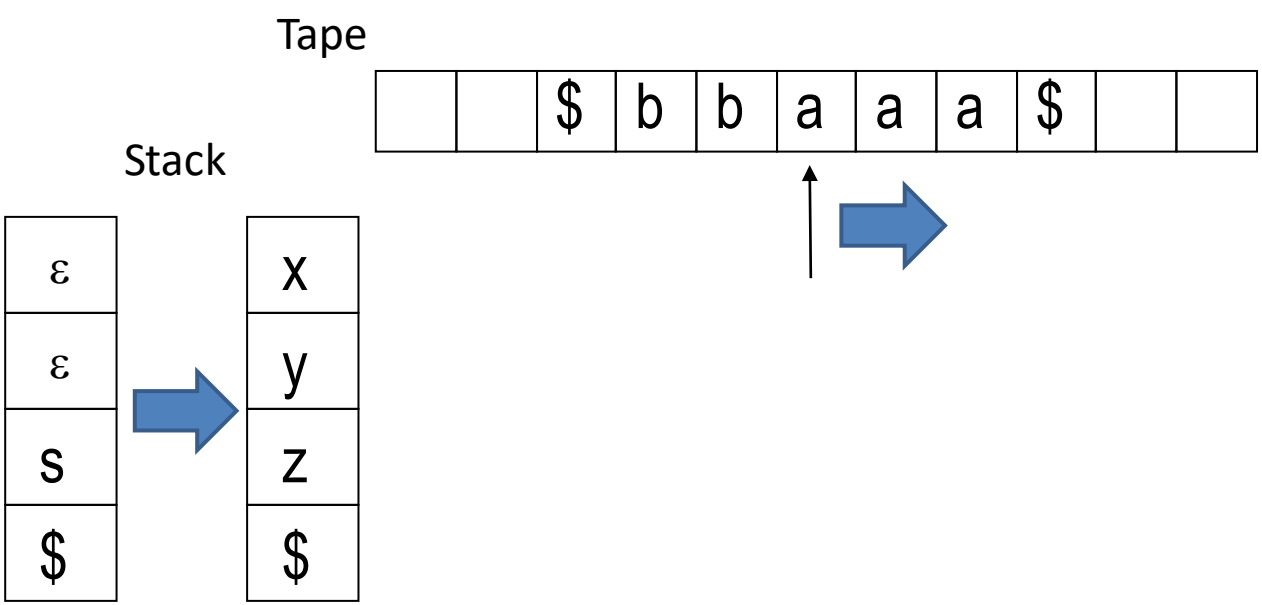


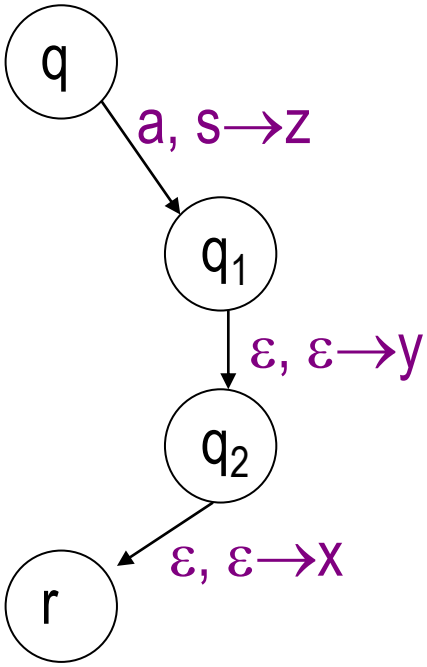
Step 2: If the top of stack is variable, then simulate the derivation

- E.g., we need derivation $S \Rightarrow xyz$

  Just Pop 'S', Push 'z','y','x'

  Input header just move by one 'a'





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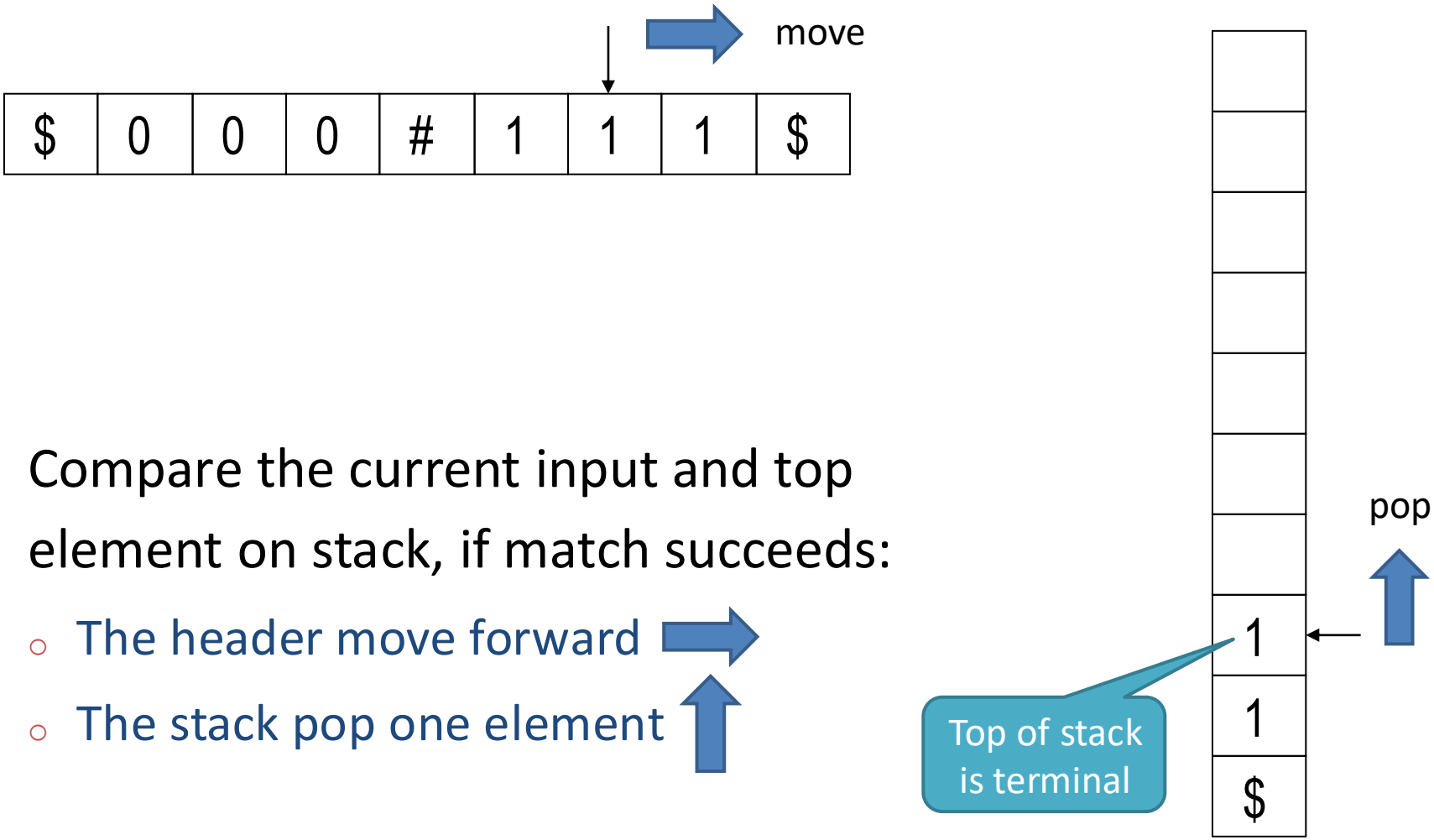

- Compare the current input and top element on stack, if match succeeds:
  - The header move forward
  - The stack pop one element

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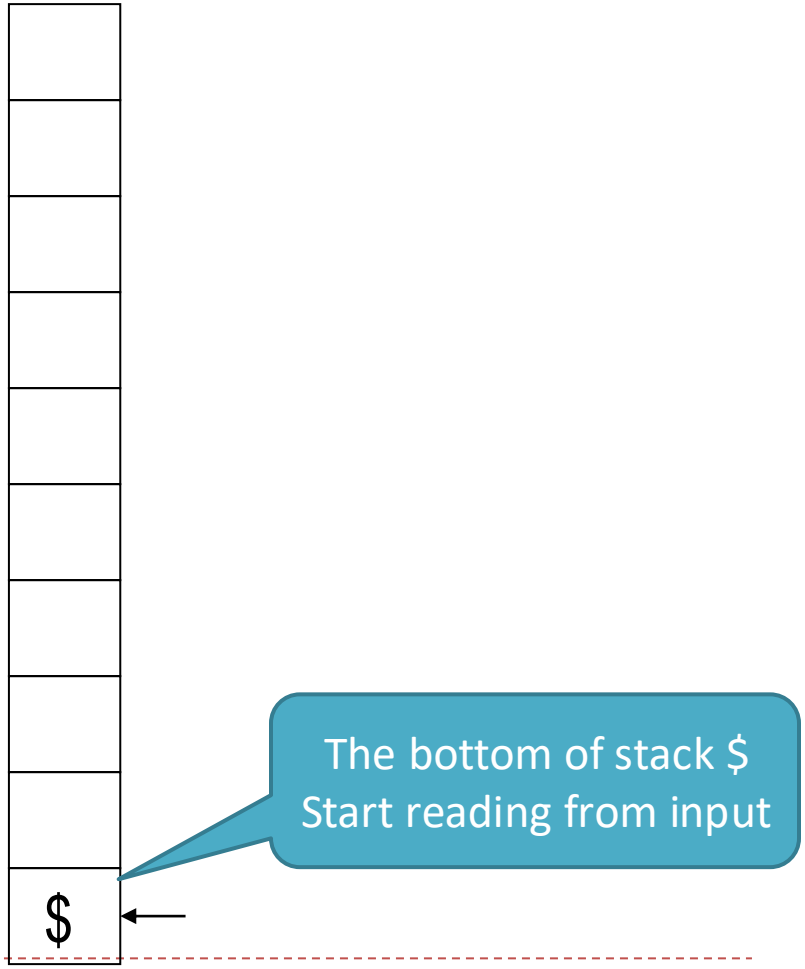

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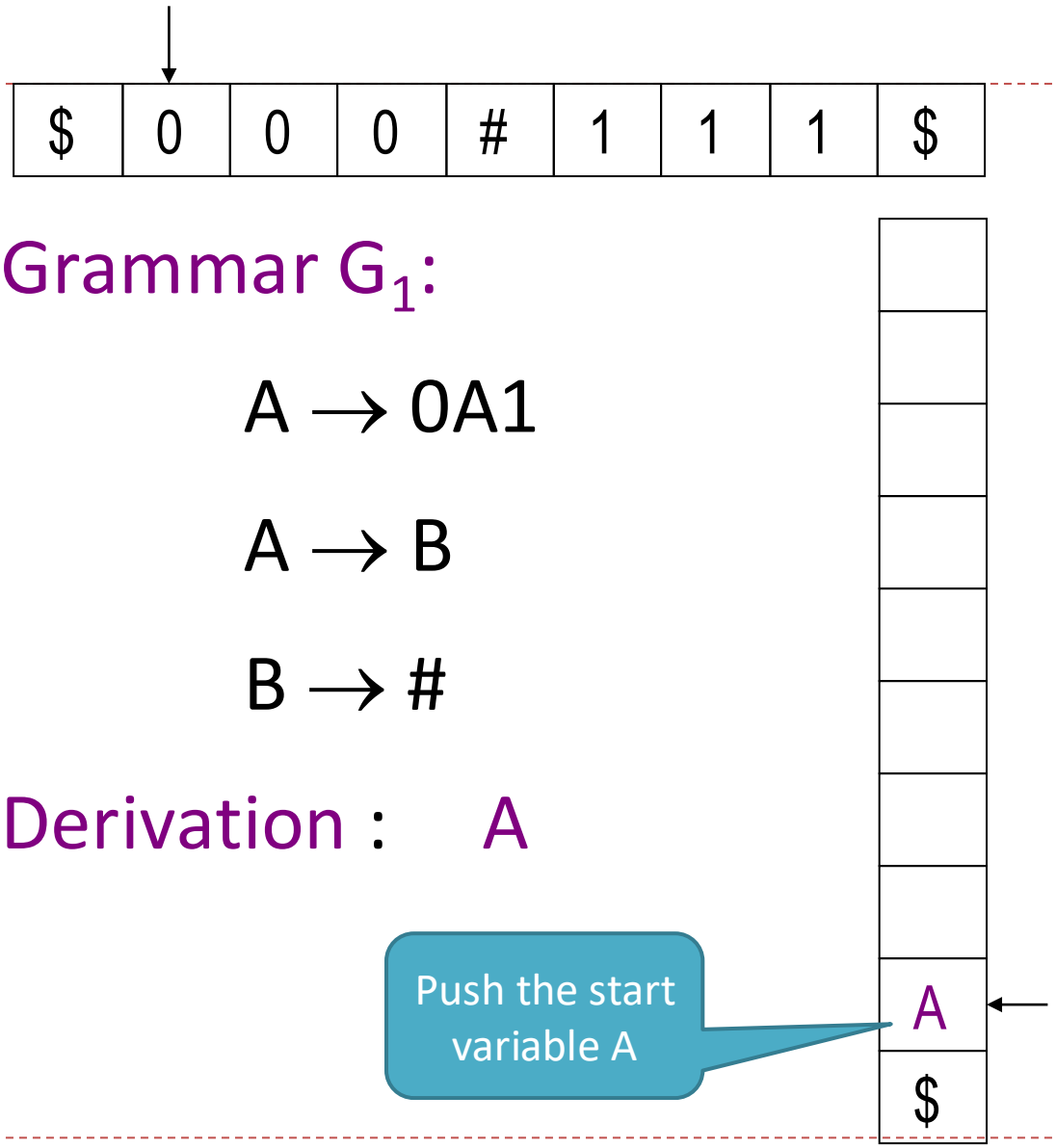

- Grammar G_1:

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- Derivation : A

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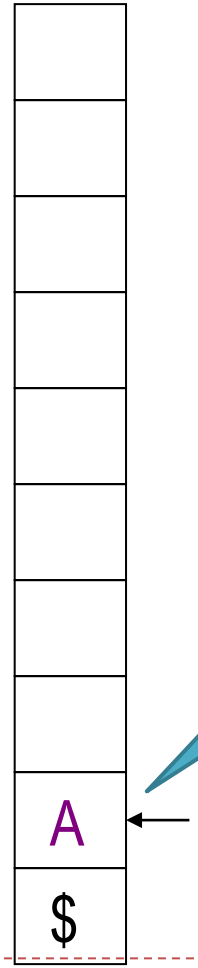

Step 2: The top of stack is variable, so simulate the derivation

Example: how PDA recognizes the input 000#111

$	0	0	0	#	1	1	1	$

During the derivation, the input is always ε; head stays still

- Gra

$A \rightarrow 0A1$

$A \rightarrow B$

$B \rightarrow \#$

- Derivation : $A \Rightarrow 0A1$

0
A
1
$

POP A
PUSH 1, A, 0

Finish the derivation:
$A \Rightarrow 0A1$

Example: how PDA recognizes the input 000#111

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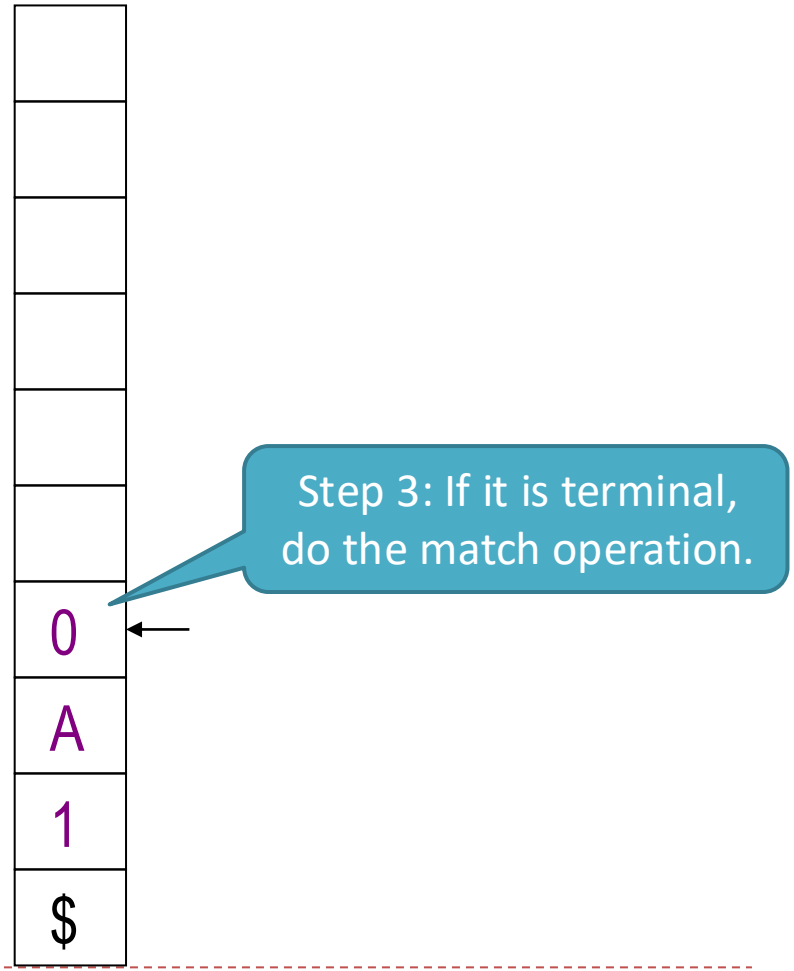

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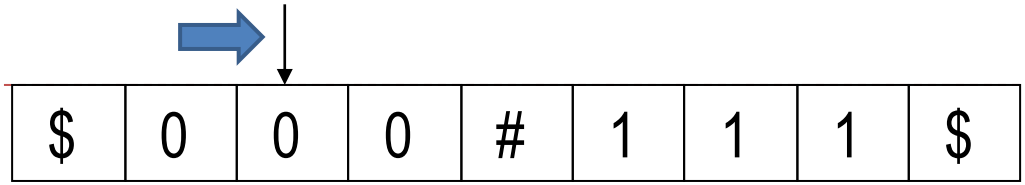

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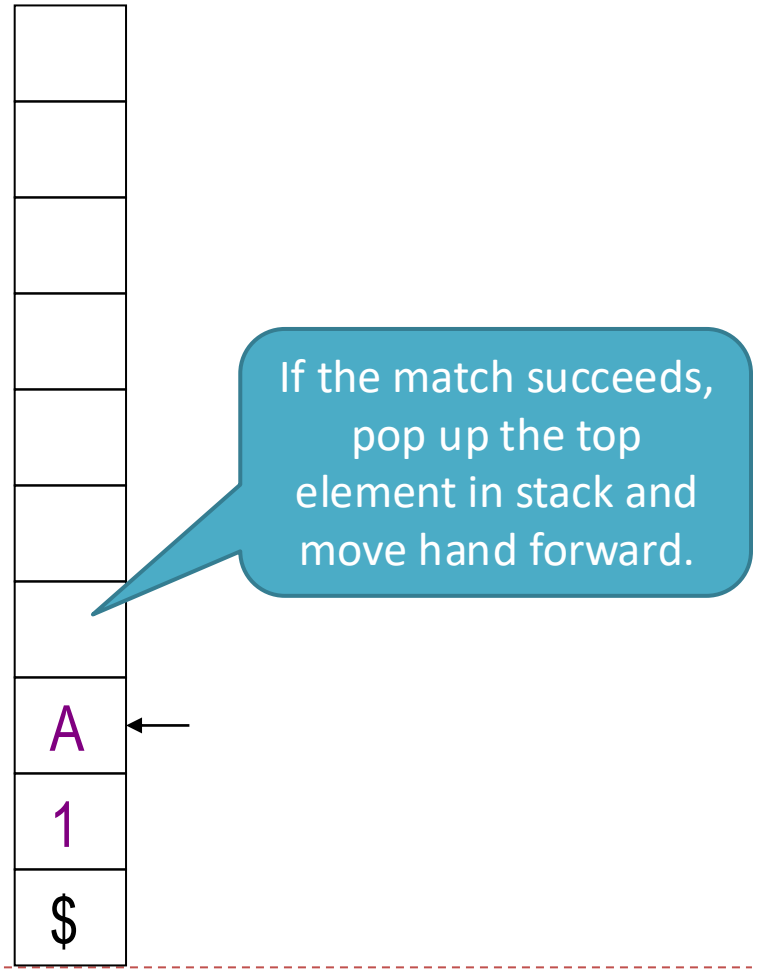

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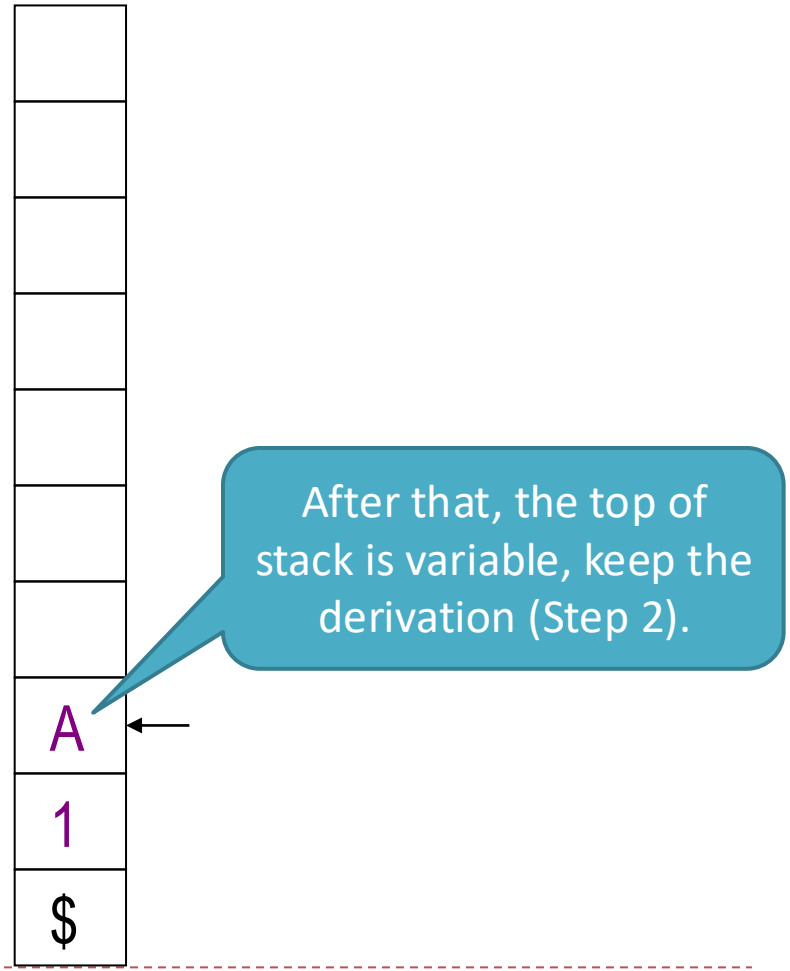

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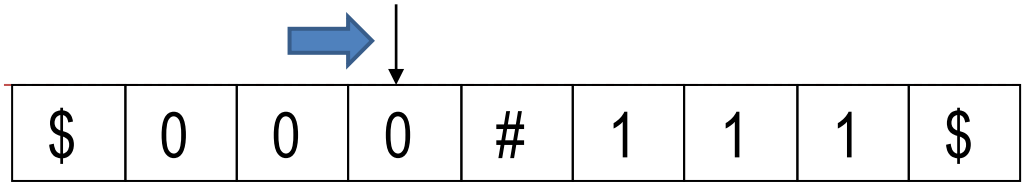
  $\Rightarrow 00A11$

0
A
1
1
$

After derivation, the top of stack becomes terminal again, keep the match with input (Step 3).

Example: how PDA recognizes the input 000#111



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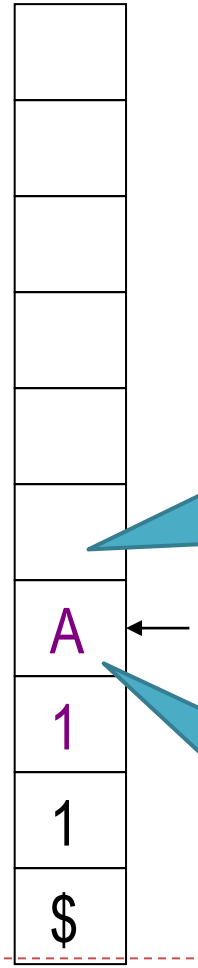

If the match succeeds, pop up the top element in stack and move hand forward.

After that, the top of stack is variable again, keep the derivation (Step 2).

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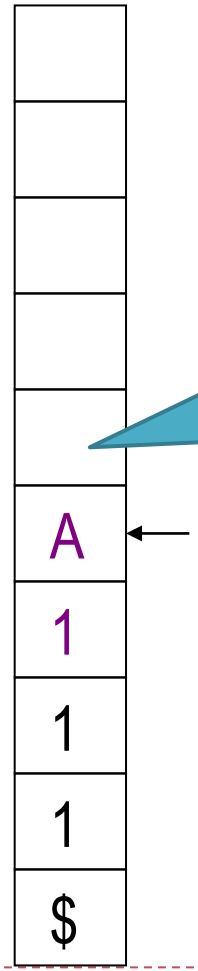

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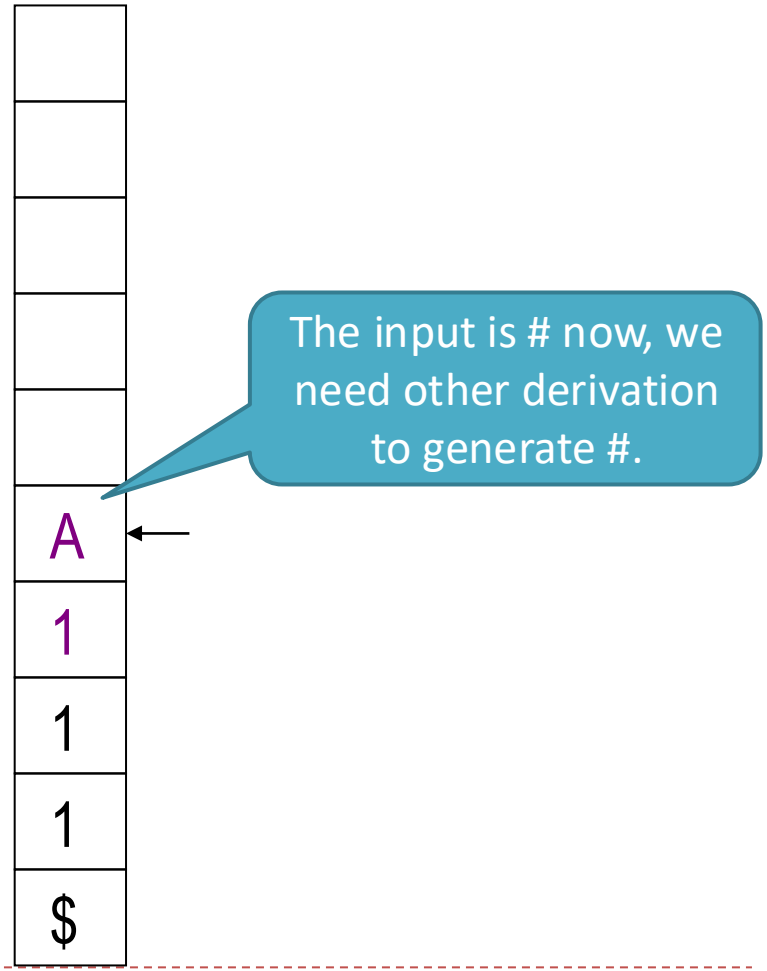

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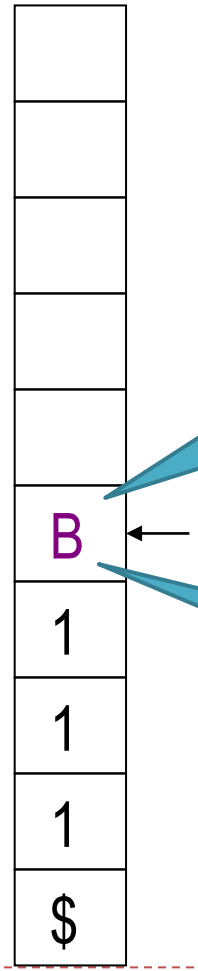

This time, we perform the derivation using grammar $A \rightarrow B$

After that, as the top element is still variable, keep the derivation (step 2)

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#
1
1
1
$

We perform the derivation using grammar $B \rightarrow \#$

The top element is terminal, then perform the match operation with input (step 3)

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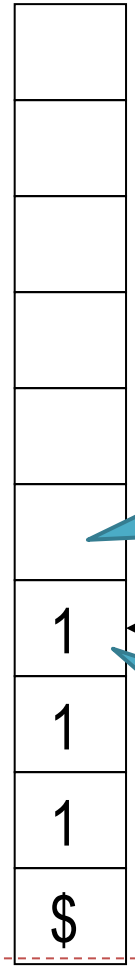

If the match succeeds, pop up the top element in stack and move hand forward.

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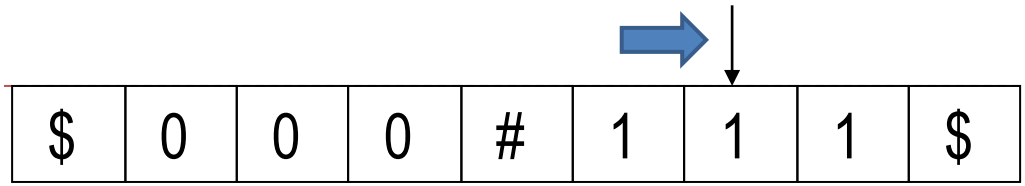

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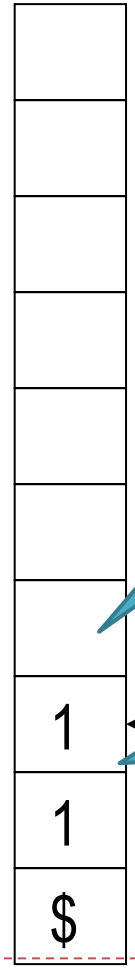

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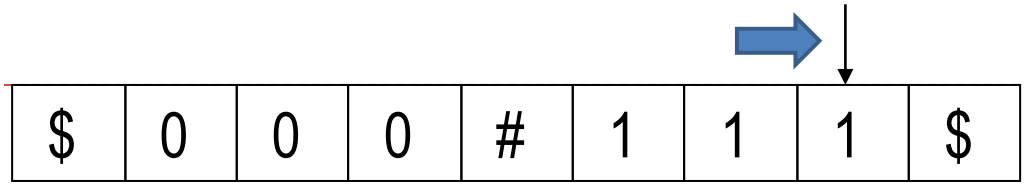

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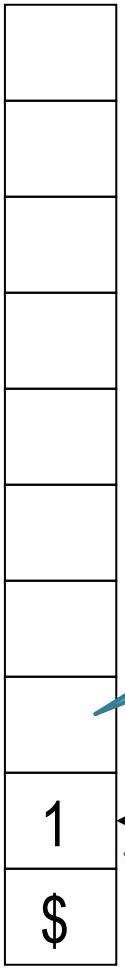

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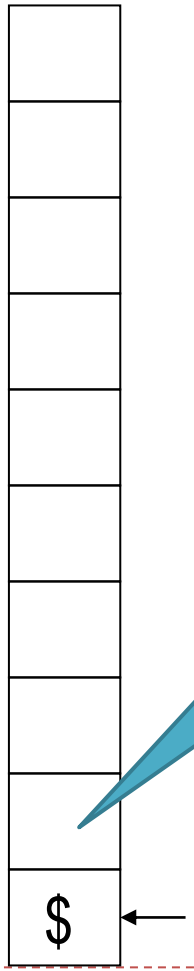

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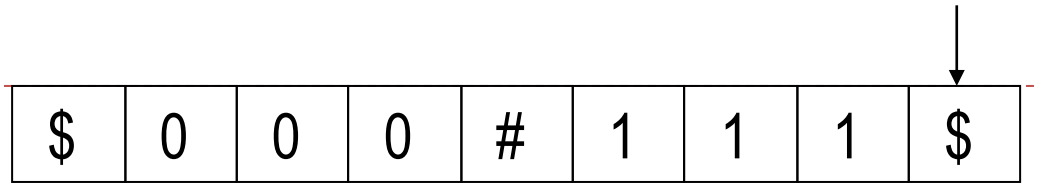

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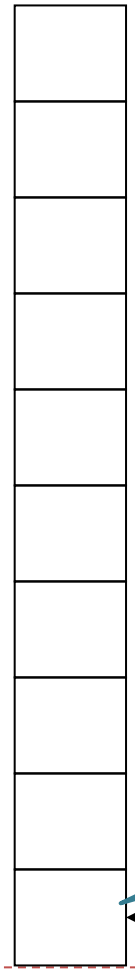

If the input is finished and stack is empty, accept; otherwise reject.

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$	a	b	a	b	$

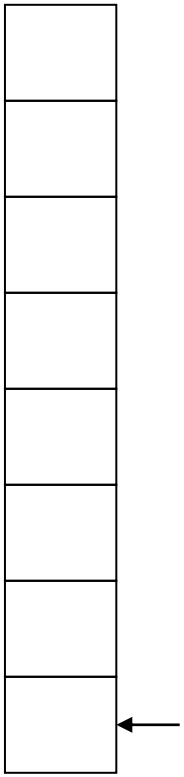
- Grammar G_3:

$$S \rightarrow aSb$$

$$S \rightarrow SS$$

$$S \rightarrow \varepsilon$$

- Derivation : $S \Rightarrow SS \Rightarrow aSbS \Rightarrow abS \Rightarrow abaSb \Rightarrow abab$



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S
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$

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S
$

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- Grammar G_3:

$$S \rightarrow aSb$$

$$S \rightarrow SS$$

$$S \rightarrow \varepsilon$$

- Derivation : $S \Rightarrow SS \Rightarrow aSbS \Rightarrow abS \Rightarrow abaSb \Rightarrow abab$

ε
b
S
$

Question

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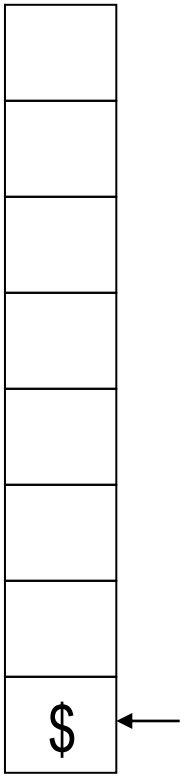

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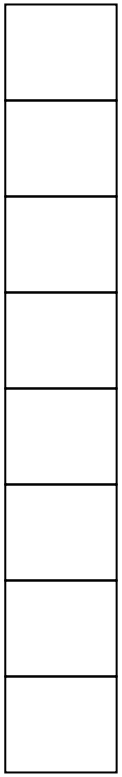

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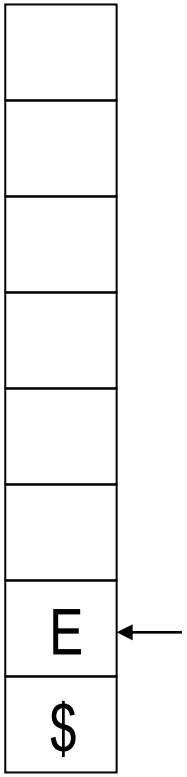

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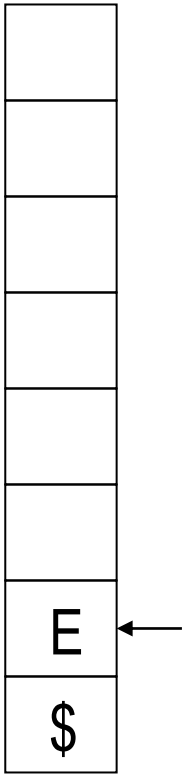

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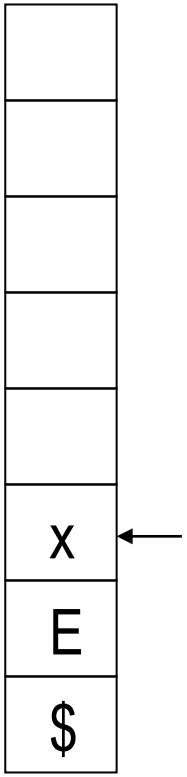

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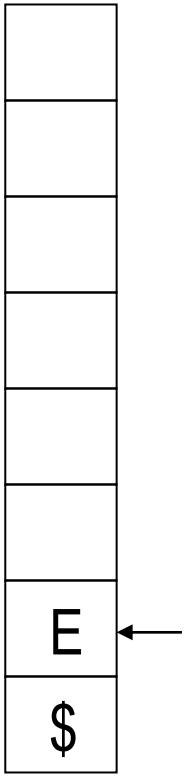

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A language is CFL $\Longrightarrow$ some PDA recognizes it

- Proof:

  Suppose A is CFL, based on definition

  we have the grammar of A, **CFG $G=(V,\Sigma,R,S)$.**

  Build the following PDA to recognize A

  $P = (\{q_{start}, q_{loop}, q_{accept}\} \cup E,$

  $\Sigma,$

  $V \cup \Sigma,$

  $\delta,$

  $q_{start},$

  $\{q_{accept}\}).$

Add states q_{start}, q_{loop}, q_{accept}
E is the intermediate state

Input is terminals in CFG

Stack could place variables and terminals in CFG

A language is CFL $\Longrightarrow$ some PDA recognizes it

Define E:

Suppose we read input **a**, state changes from **q** to **r**, and the top of stack changes from **s** to $\mathbf{u_1u_2\ldots u_k}$

introduce new states $q_1,q_2,\ldots,q_{k-1}$,

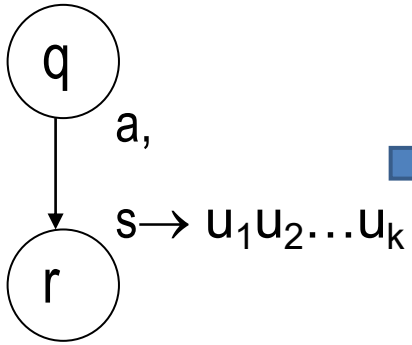
let $\delta(q,a,s)$ contains (q_1,u_k),

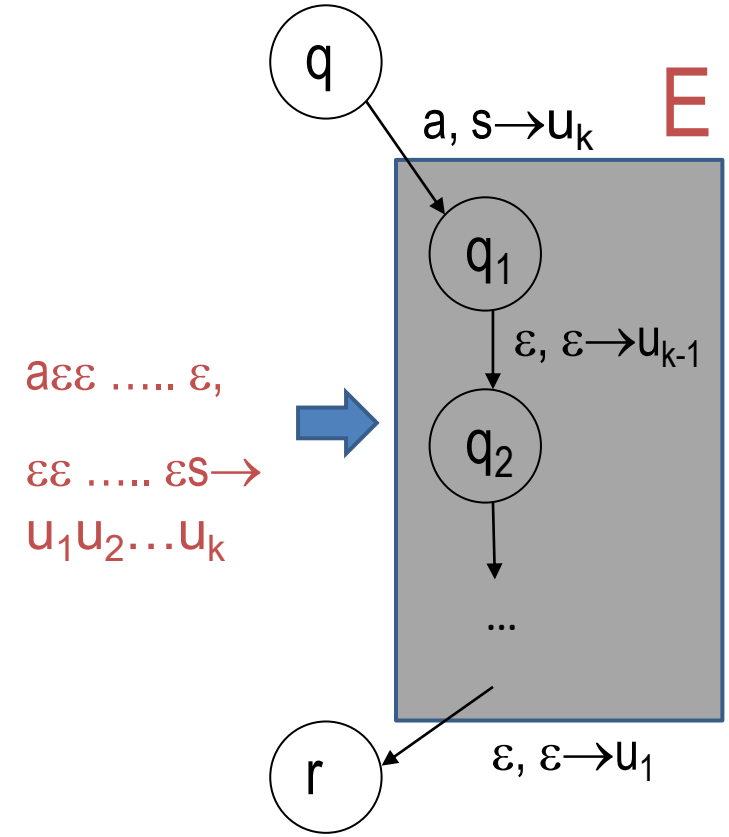
$\delta(q_1,\varepsilon,\varepsilon)=\{(q_2,u_{k-1})\}$,

$\delta(q_2,\varepsilon,\varepsilon)=\{(q_3,u_{k-2})\}$,

…,

$\delta(q_{k-1},\varepsilon,\varepsilon)=\{(r,u_1)\}$.





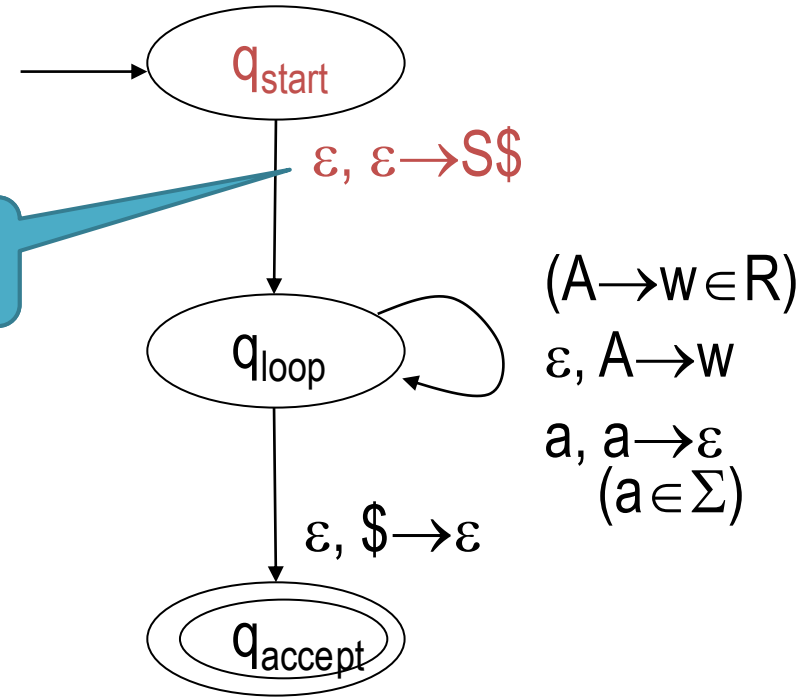
Conclude all the above as $(r,u)\in\delta(q,a,s)$ and mark all the new states as E

A language is CFL $\Longrightarrow$ some PDA recognizes it

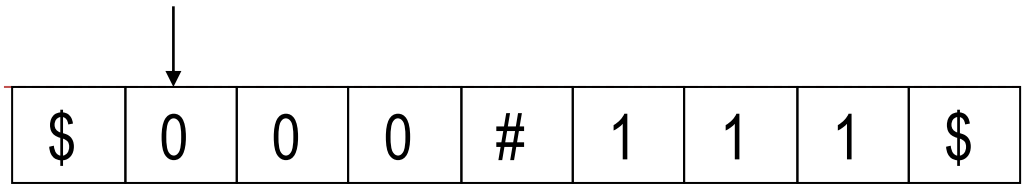
Define δ :

$\delta(q_{start},\varepsilon,\varepsilon)=\{(q_{loop},S\$)\}$,

Put $ and start variable into stack



Example: Put $ and start variable into stack



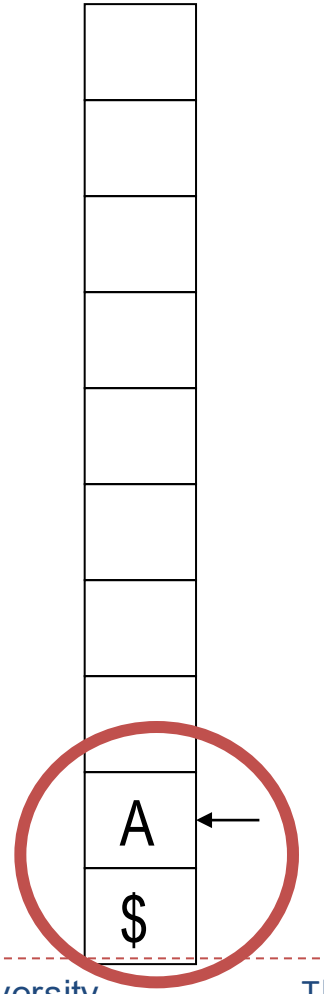
- Grammar G_1:

  $A \rightarrow 0A1$

  $A \rightarrow B$

  $B \rightarrow \#$

- Derivation : A



ge is CFL $\Longrightarrow$ some PDA recognizes it

top of stack is variable A

replace A by w

pick up one rule $A \to w$

Define δ

$\delta(q_{loop},\varepsilon,A)=\{ (q_{loop},w) \mid A\to w \text{ is rule of } R\}$,

$\delta(q_{loop},a,a)=\{(q_{loop},\varepsilon)\}$,

top of stack is terminal a

Match succeeds and pop

// repeat the following steps

If the top of stack is variable A,

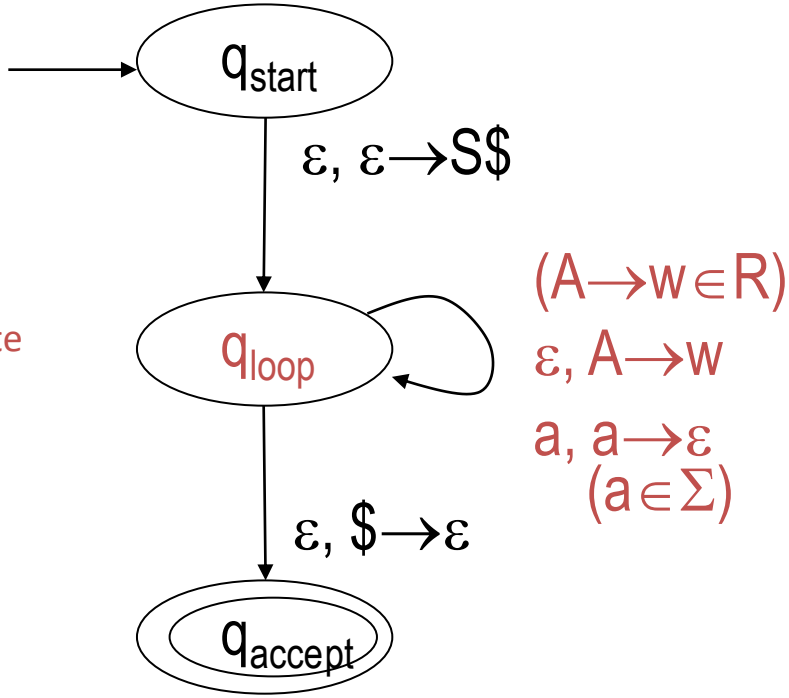
then pick up one rule $A\to w$, and replace A by w, simulate the derivation (Step 2).

If the top of stack is terminal a,

then compare a with the symbol of current input

if match(Step 3), repeat;

if not match, reject this branch.



A language is CFL $\Longrightarrow$ some PDA recognizes it

top of stack is $

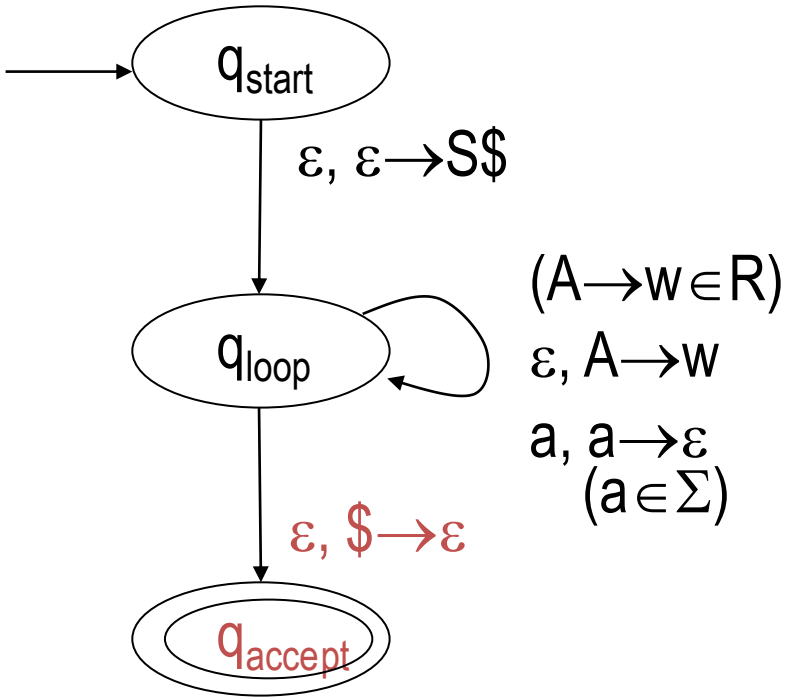
the input is over

Define δ :

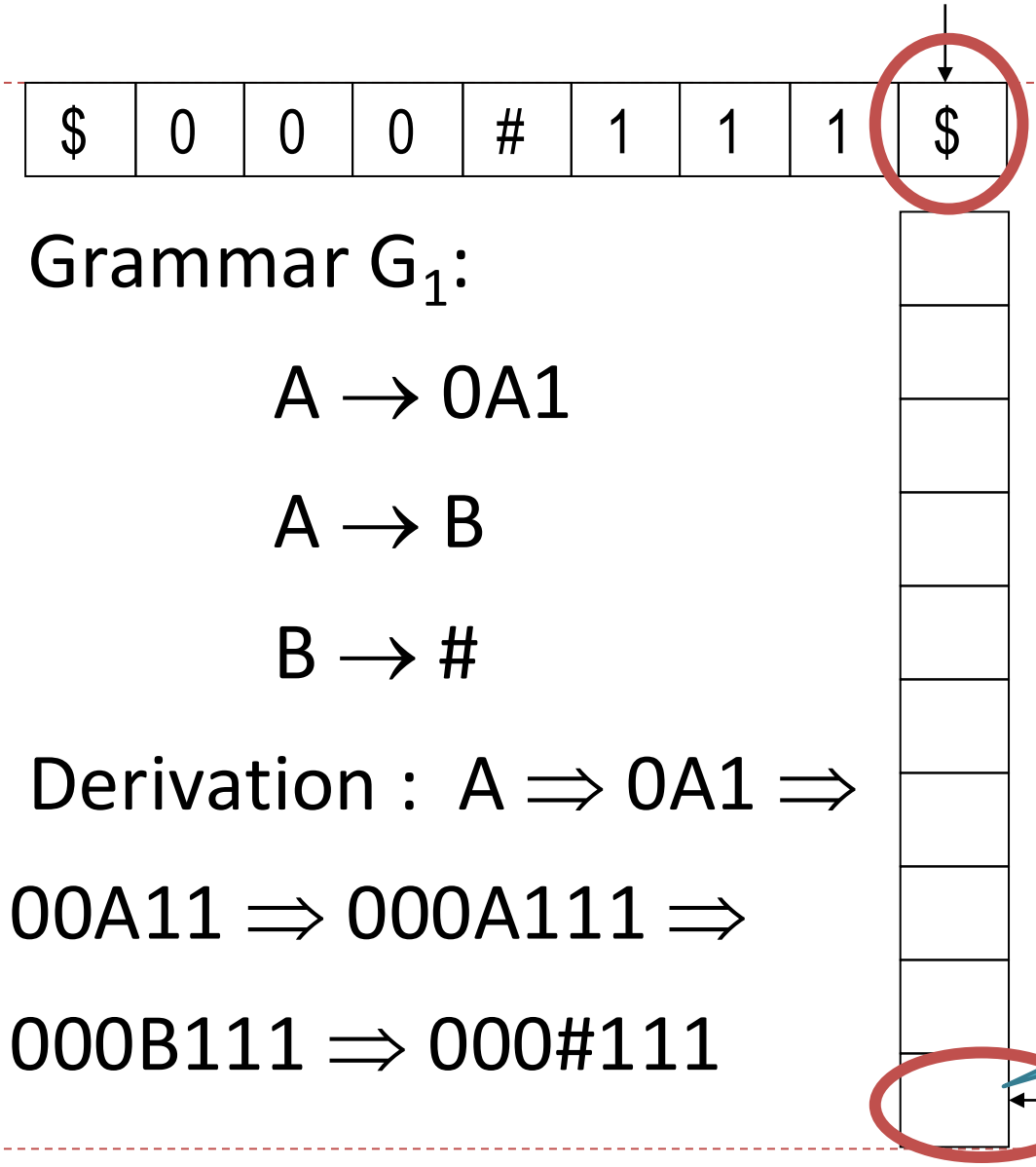
$\delta(q_{loop},\varepsilon,\$)=\{(q_{accept},\varepsilon)\}$.

// If the top of stack is $,

if the input is over, accept.



Example: when stack and input are empty



- Grammar G_1:

$$A \rightarrow 0A1$$

$$A \rightarrow B$$

$$B \rightarrow \#$$

- Derivation : $A \Rightarrow 0A1 \Rightarrow$

$00A11 \Rightarrow 000A111 \Rightarrow$

$000B111 \Rightarrow 000\#111$

If the input is finished and stack is empty, accept; otherwise reject.

A language is CFL $\Longrightarrow$ some PDA recognizes it

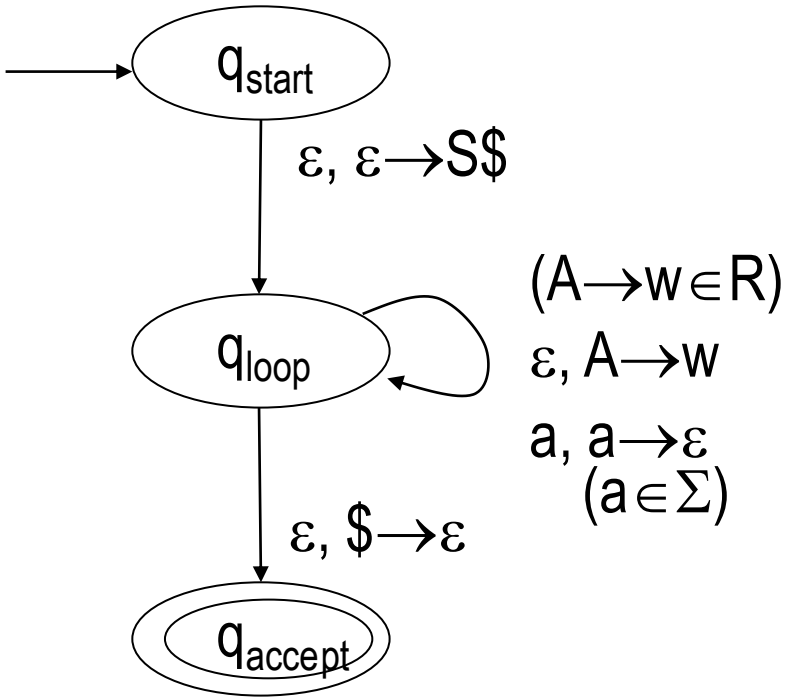
Define δ :

$\delta(q_{start},\varepsilon,\varepsilon)=\{(q_{loop},S\$)\}$, //step 1

$\delta(q_{loop},\varepsilon,A)=\{ (q_{loop},w) \mid A\rightarrow w \text{ is rule of } R\}$,

$\delta(q_{loop},a,a)=\{(q_{loop},\varepsilon)\}$, //step 2 or 3

$\delta(q_{loop},\varepsilon,\$)=\{(q_{accept},\varepsilon)\}$.



Conclusion for A language is CFL $\Longrightarrow$ some PDA recognizes it

Put $ and start variable into stack

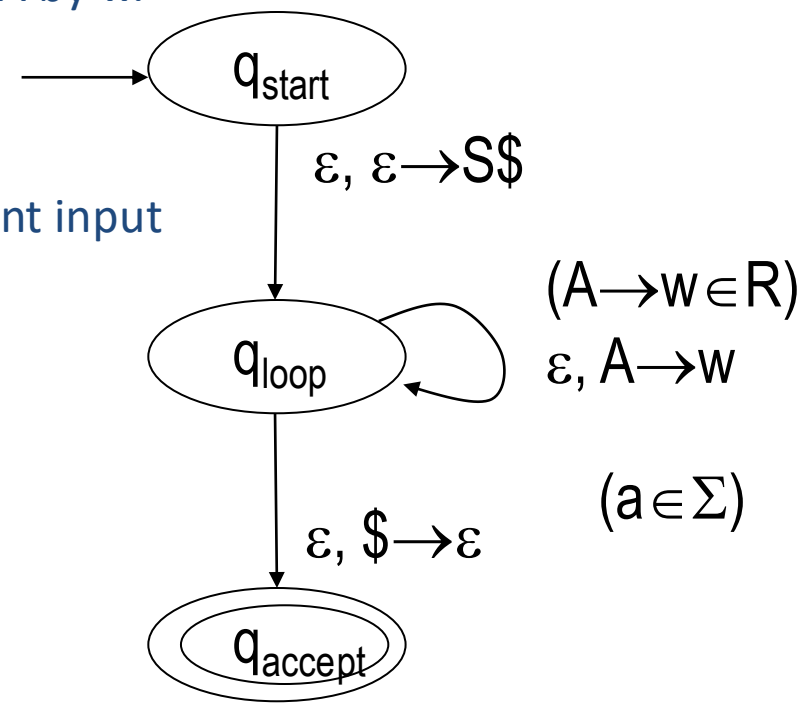
Repeat the following steps

- If the top of stack is variable A,
  then pick up one rule $A \rightarrow w$, and replace A by w.
- If the top of stack is terminal a,
  then compare a with the symbol of current input
  - if match, repeat;
  - if not match, reject this branch.

If the top of stack is $,

if the input is over, accept.

Create a PDA like this:



Example of CFL $\Longrightarrow$ PDA

- CFG G: $S \rightarrow aTb$, $T \rightarrow Ta$, $S \rightarrow b$, $T \rightarrow \varepsilon$

  construct an equivalent PDA P_1.

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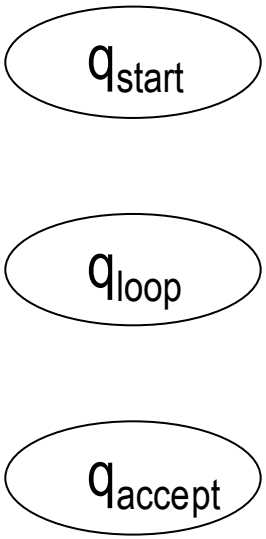

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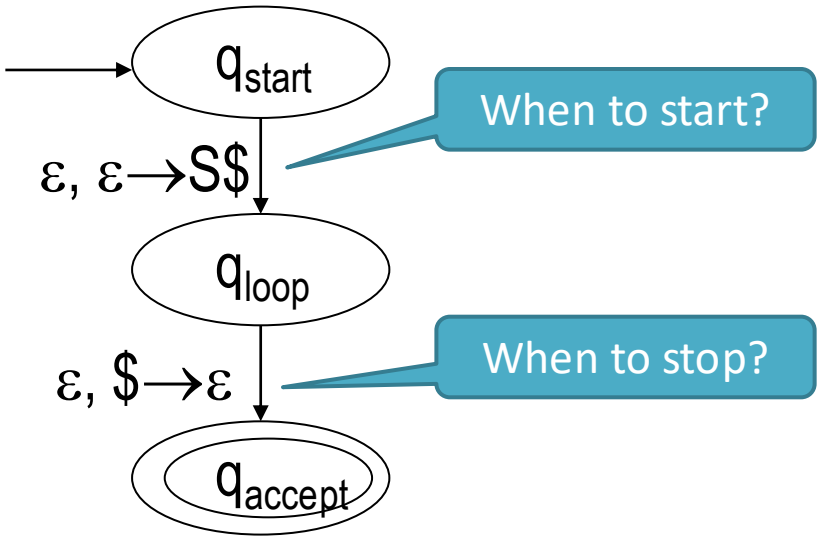

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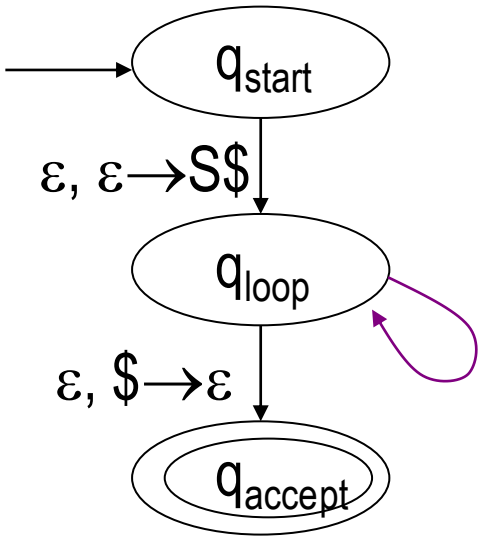

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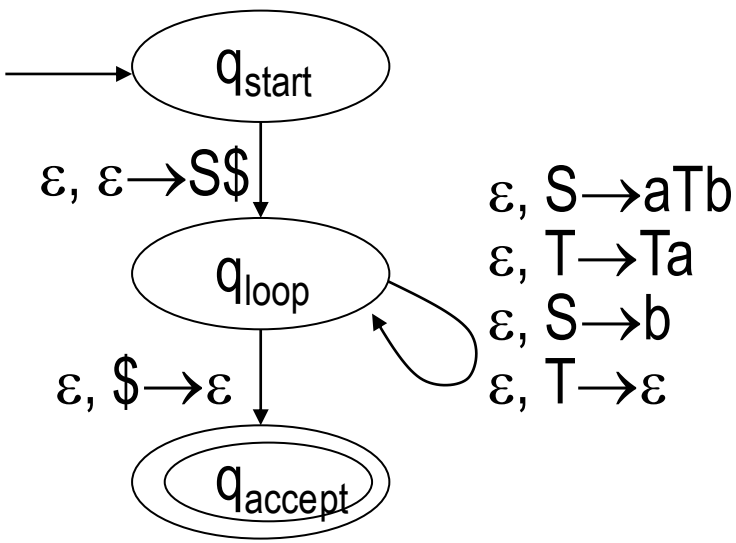

STEP 2: simulate the derivation

4 rules --> 4 transition functions

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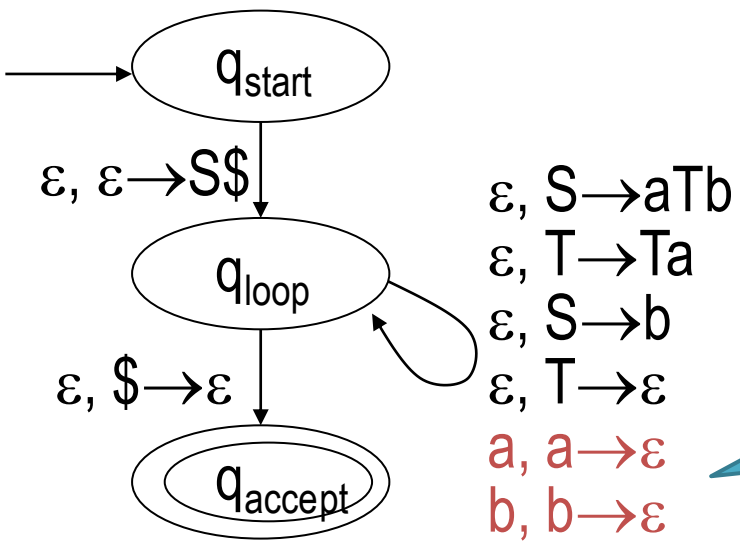

STEP 3: do the match

2 matches: input head move forward and terminal pop out

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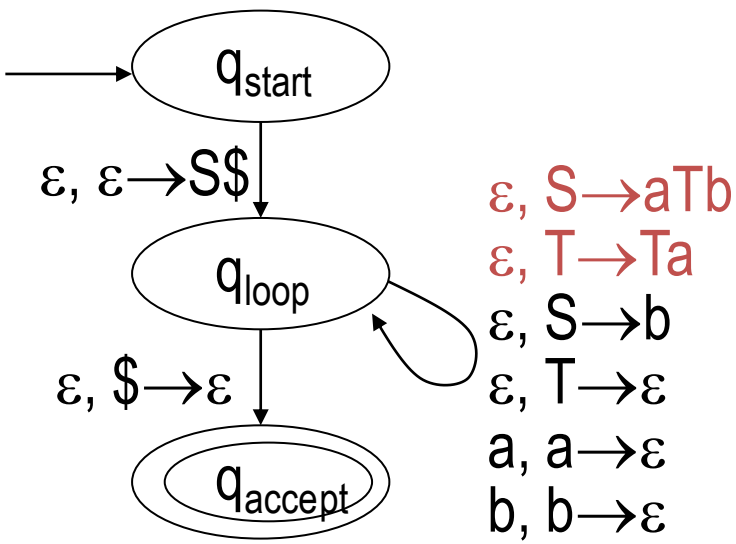


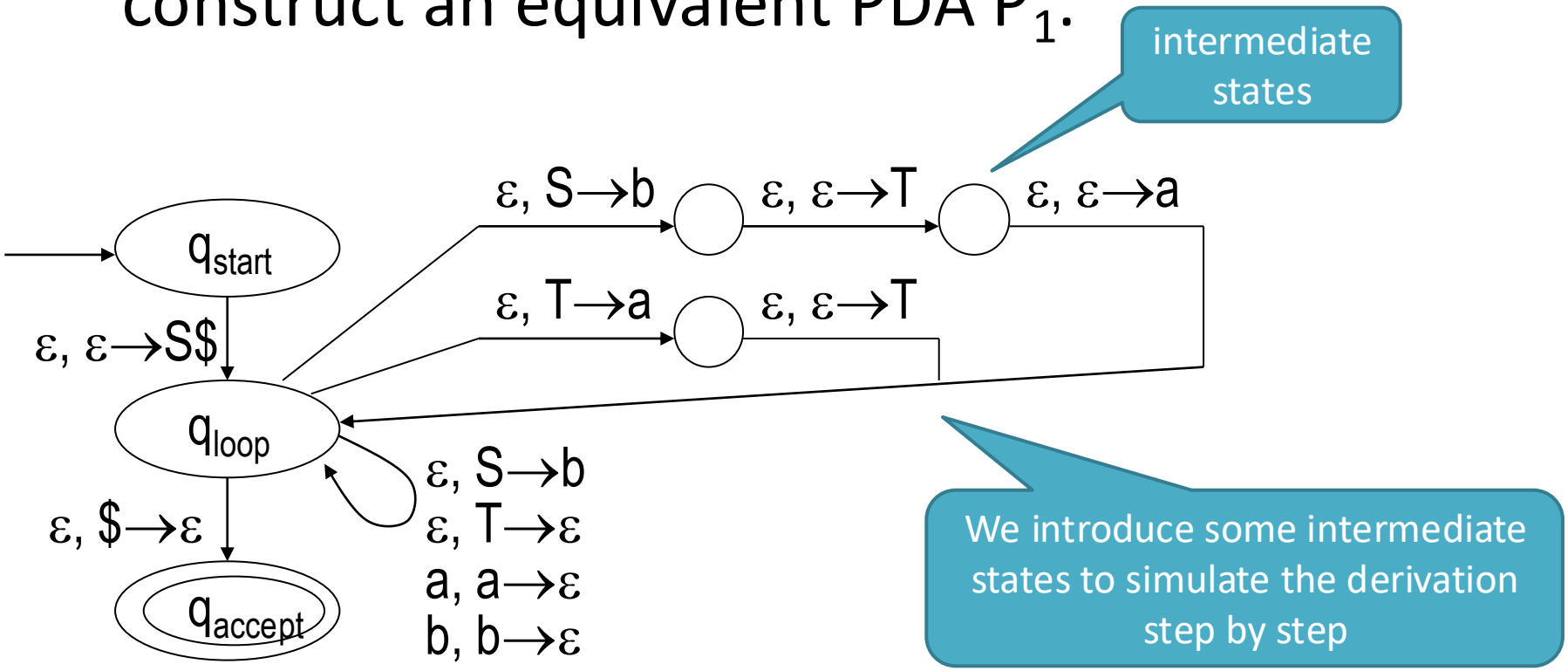
$\varepsilon, S \rightarrow aTb$
$\varepsilon, T \rightarrow Ta$
$\varepsilon, S \rightarrow b$
$\varepsilon, T \rightarrow \varepsilon$
$a, a \rightarrow \varepsilon$
$b, b \rightarrow \varepsilon$

For these which right side has one more symbols, we introduce some intermediate states

Example of CFL $\Longrightarrow$ PDA

- CFG G: $S \to aTb$, $T \to Ta$, $S \to b$, $T \to \varepsilon$

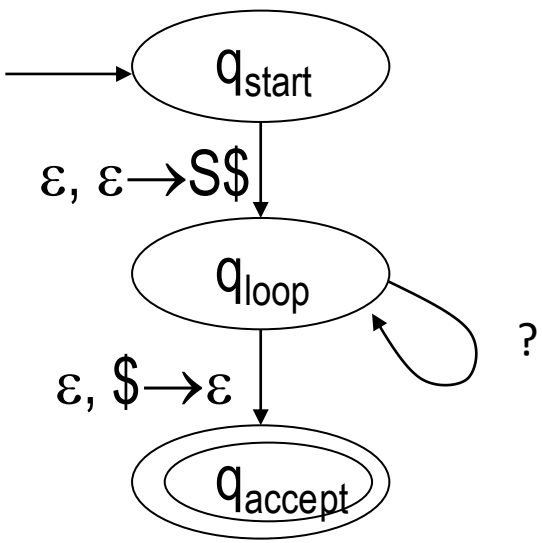
  construct an equivalent PDA P_1.



Question

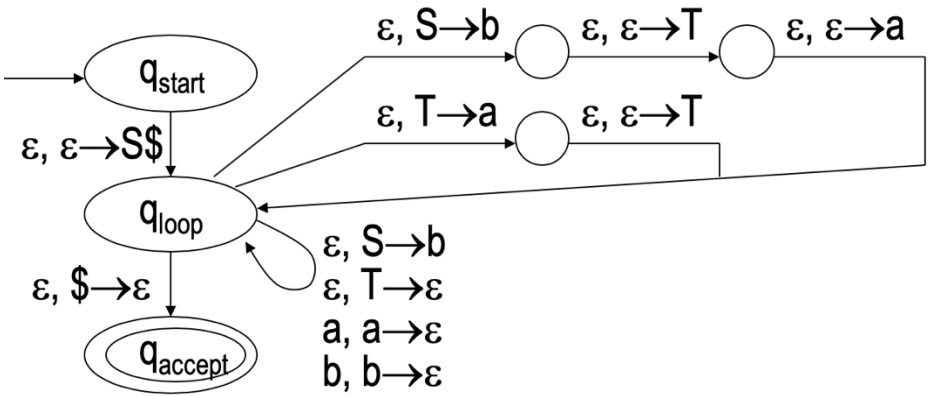
- CFG G: $E \rightarrow E+E \mid E\times E \mid (E) \mid a$

  construct an equivalent PDA P_2.



- CFG G: $S\rightarrow aTb$, $T\rightarrow Ta$, $S\rightarrow b$, $T\rightarrow \varepsilon$

  construct an equivalent PDA P_1.



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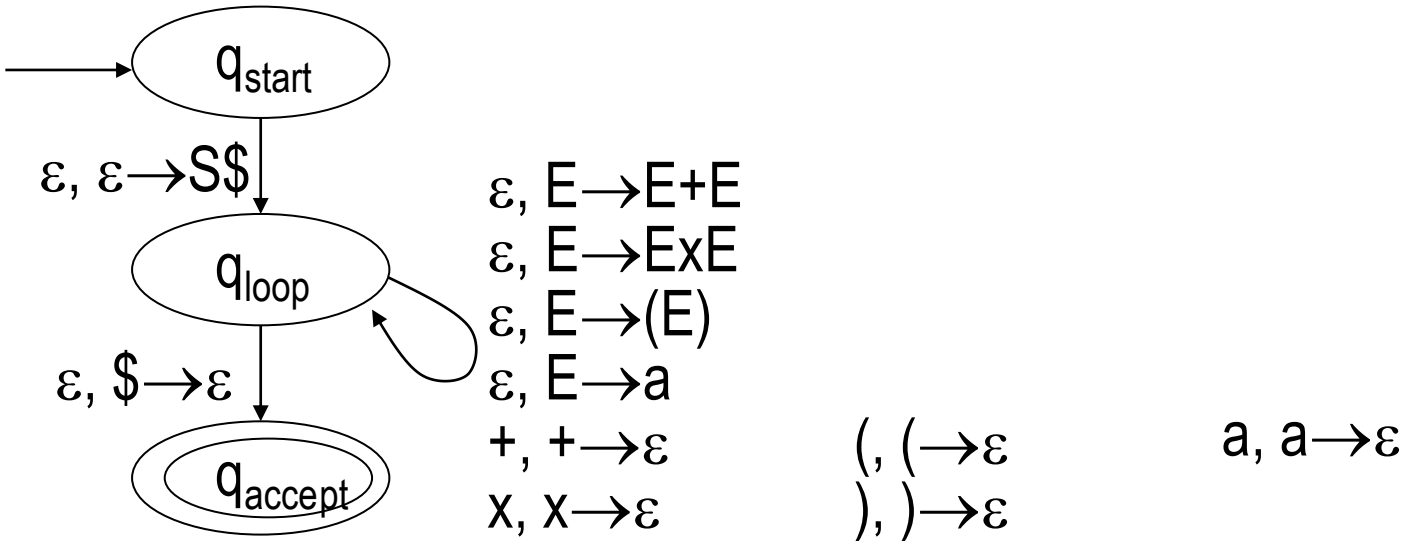

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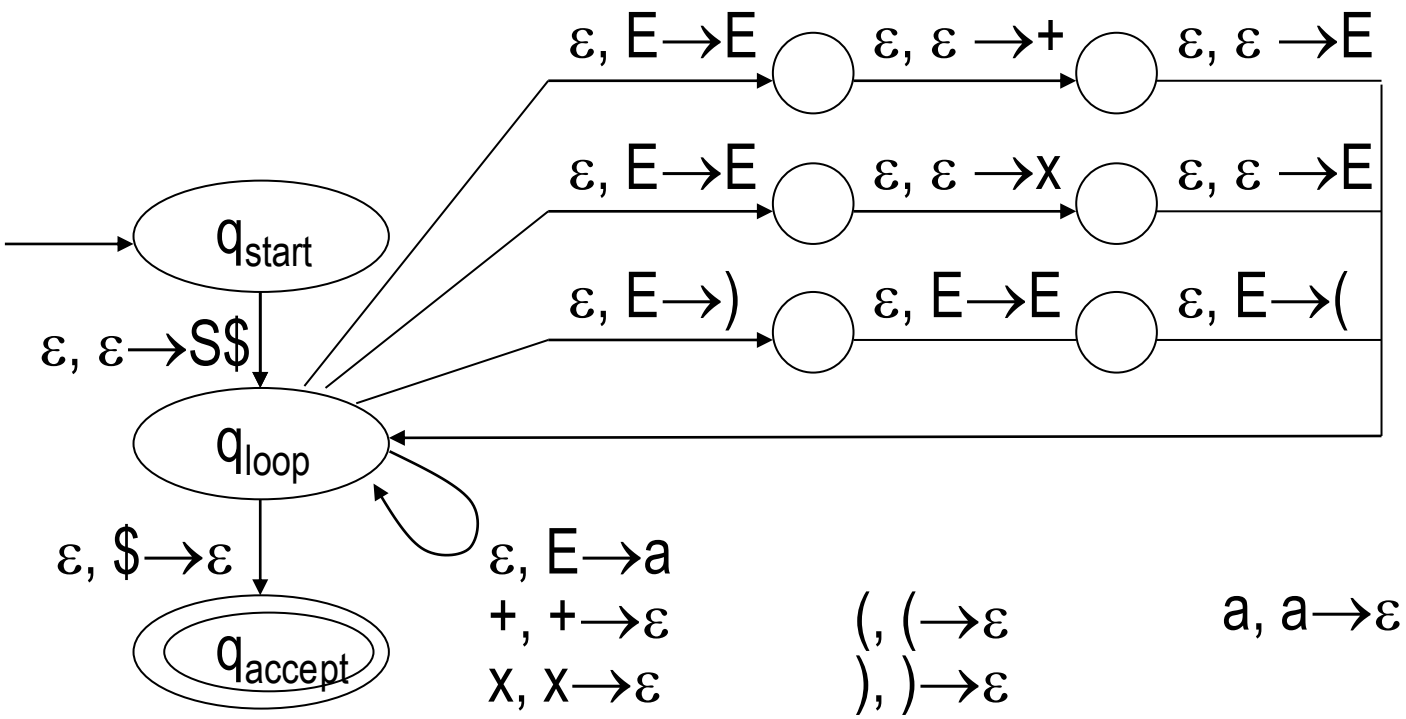


Outline

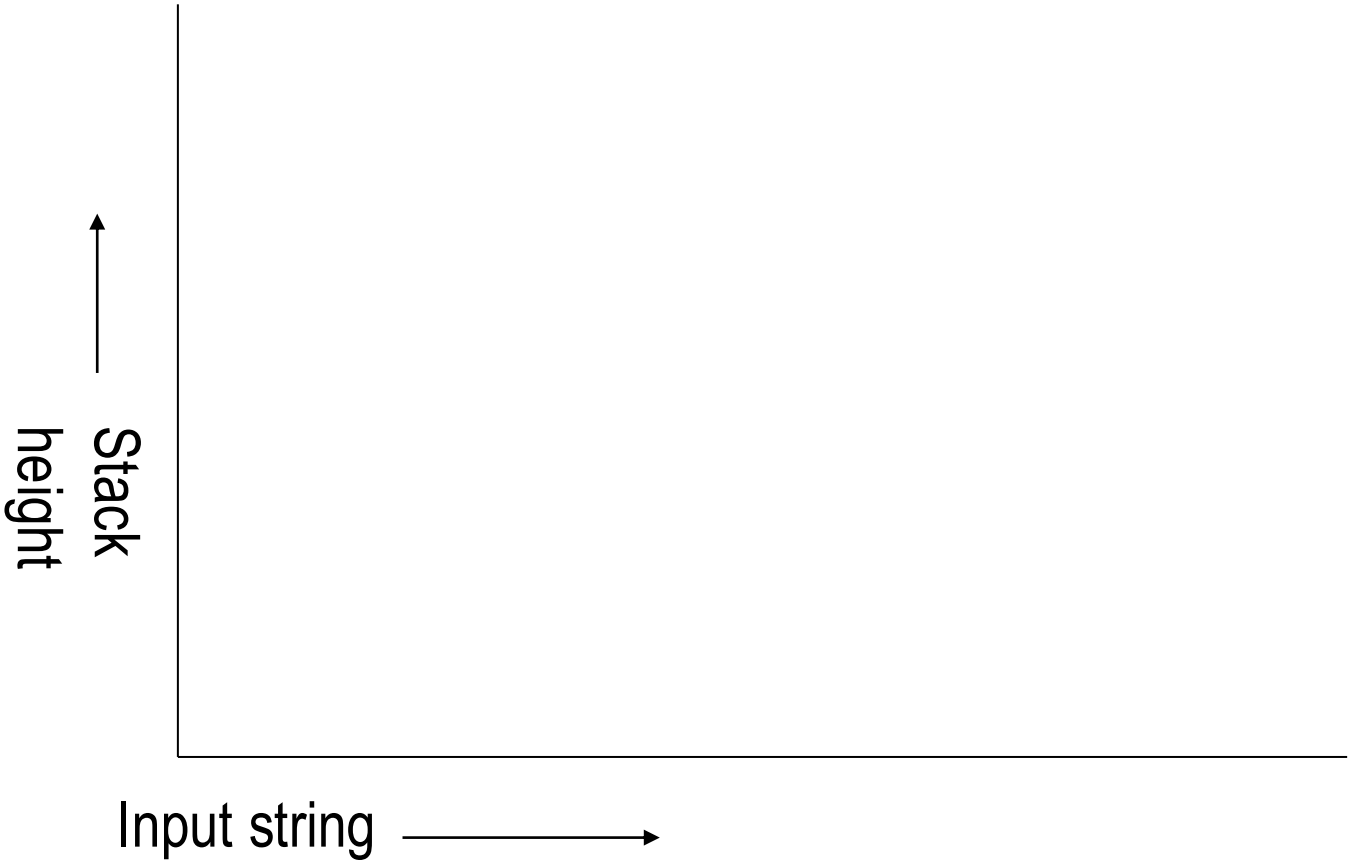
- Equivalence of PDA and CFG:

- A language is CFL $\Longrightarrow$ some PDA recognizes it

- **A language is CFL $\Longleftarrow$ some PDA recognizes it**

A language is CFL ⇐ some PDA recognizes it

- The process of PDA recognizes some inputs



A language is CFL ⇐ some PDA recognizes it

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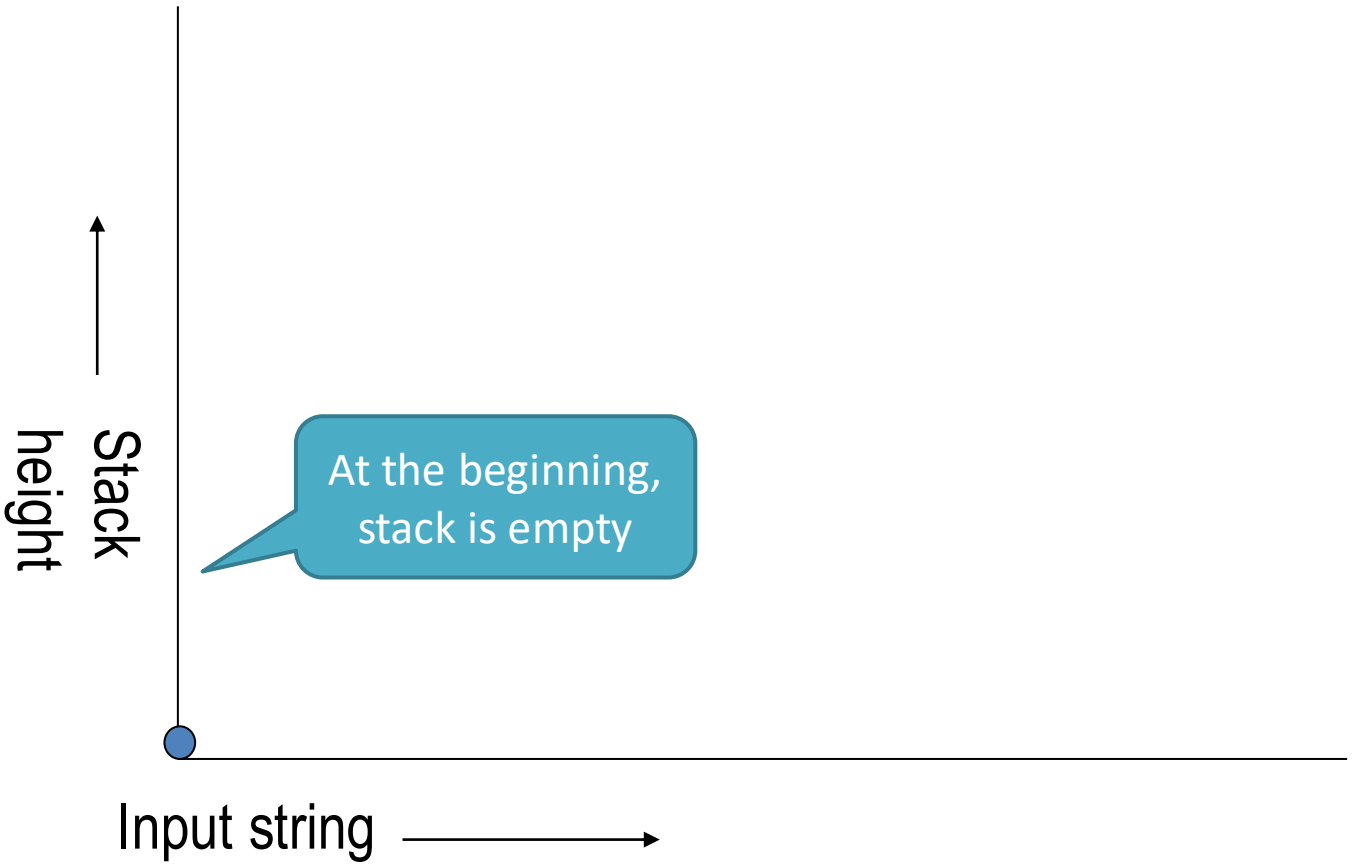

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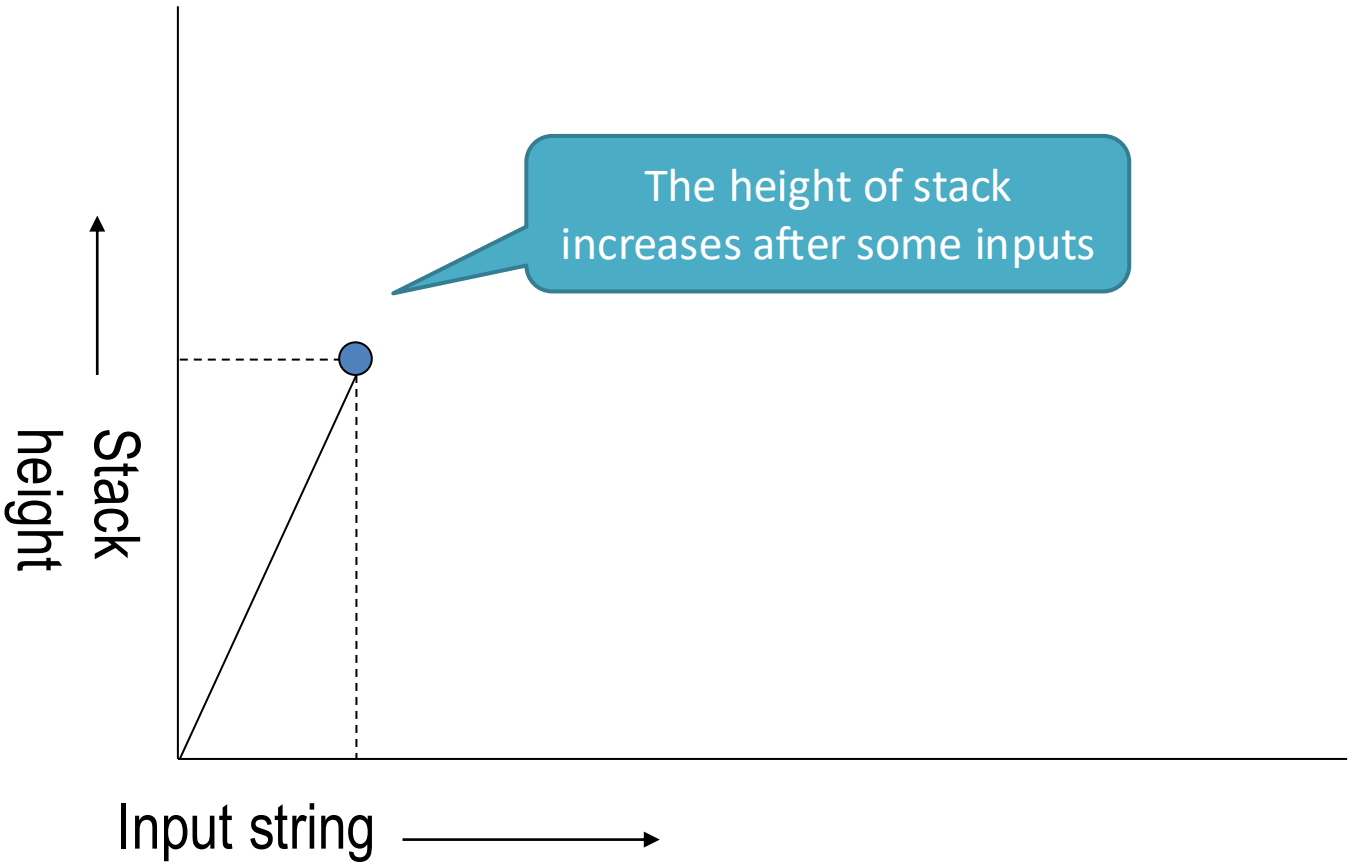
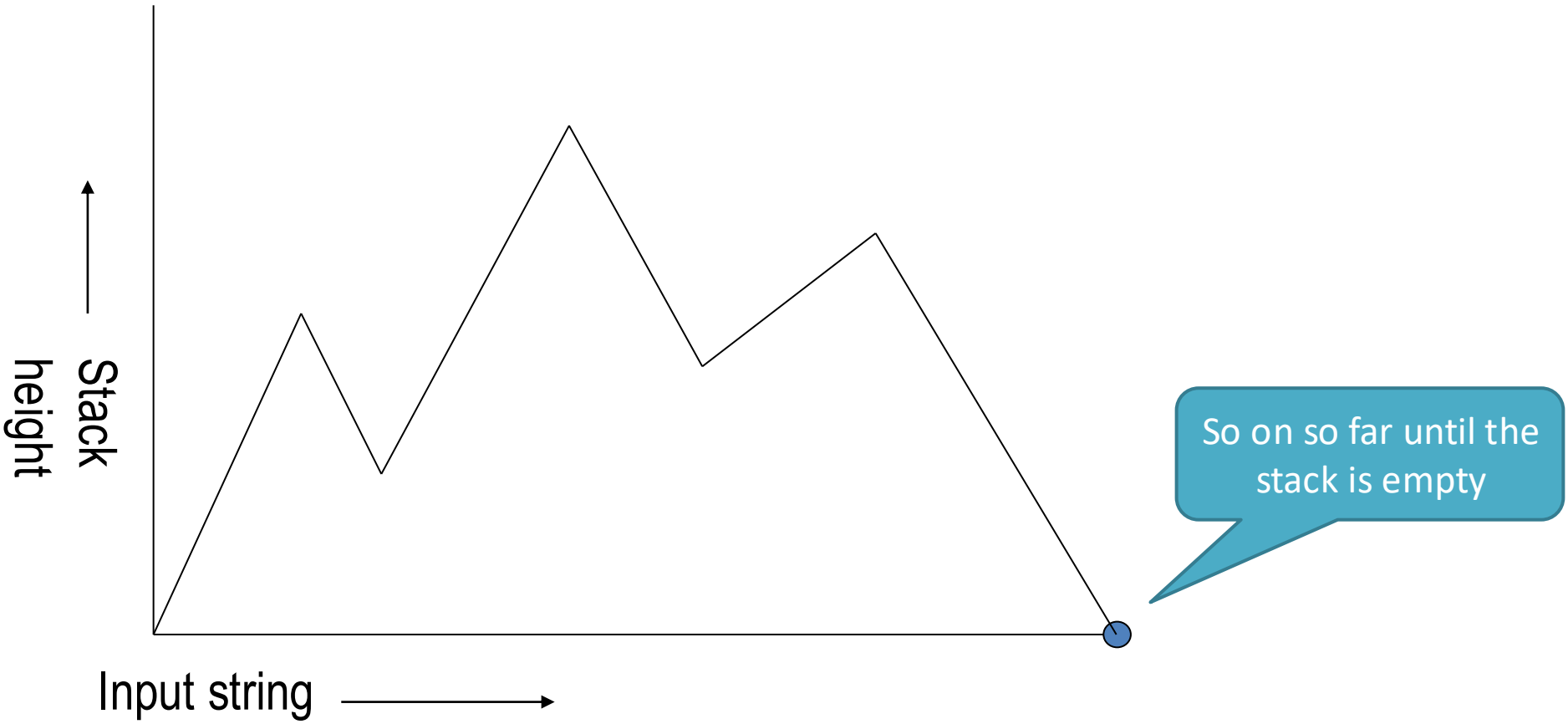


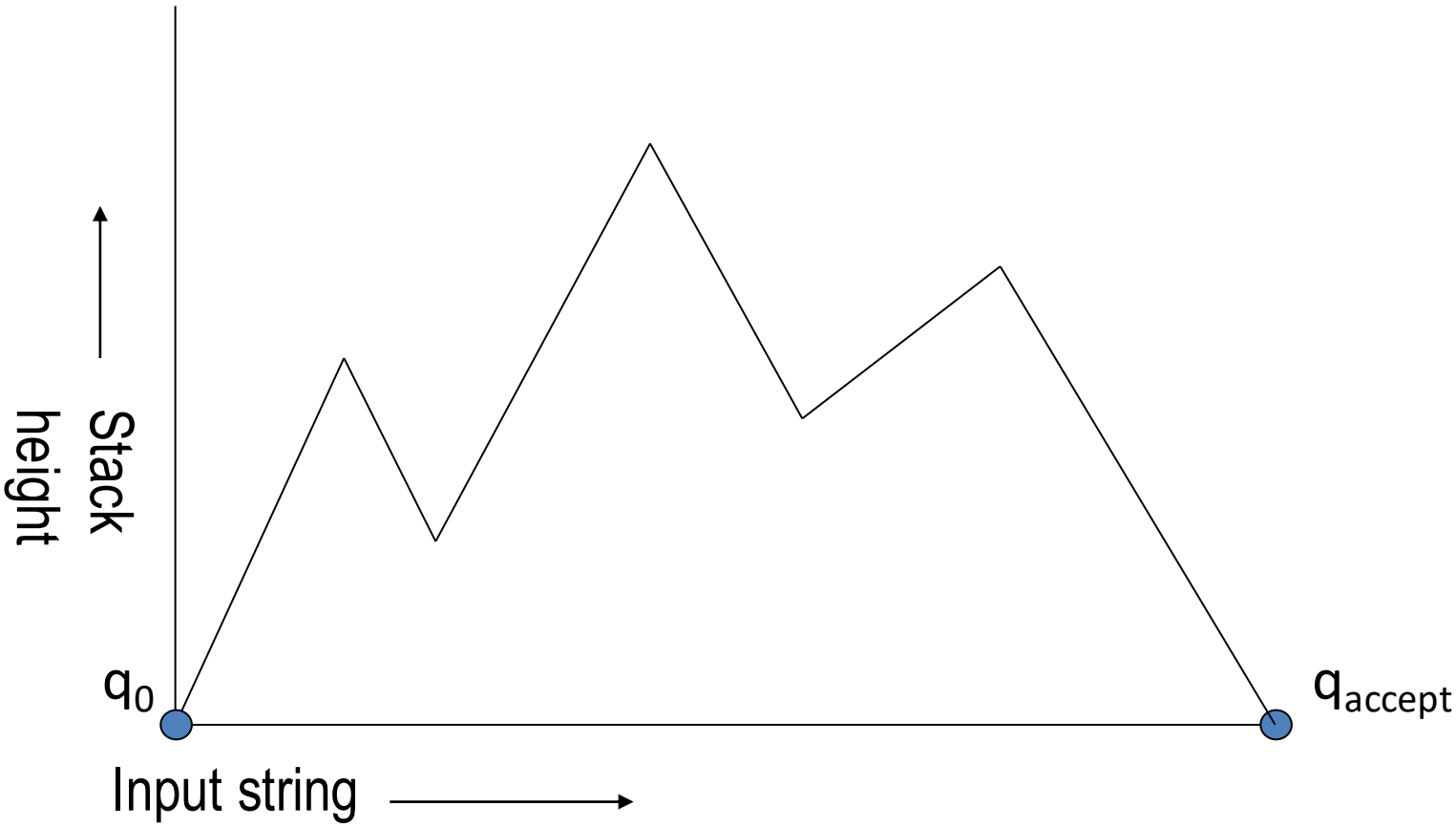
A language is CFL ⇐ some PDA recognizes it

- The process of PDA recognizes some inputs



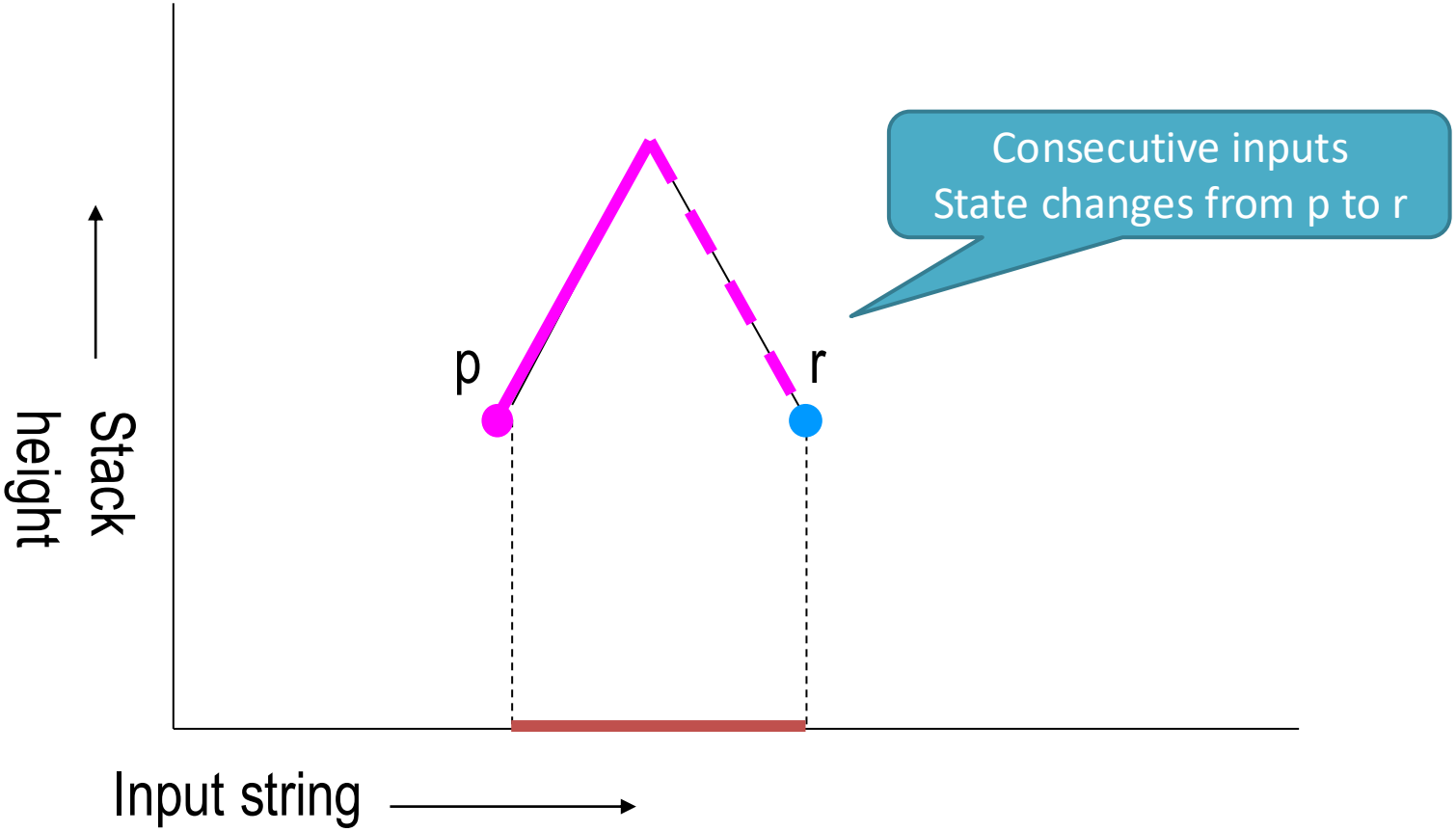
A language is CFL ⇐ some PDA recognizes it

- Create CFGs for such movements



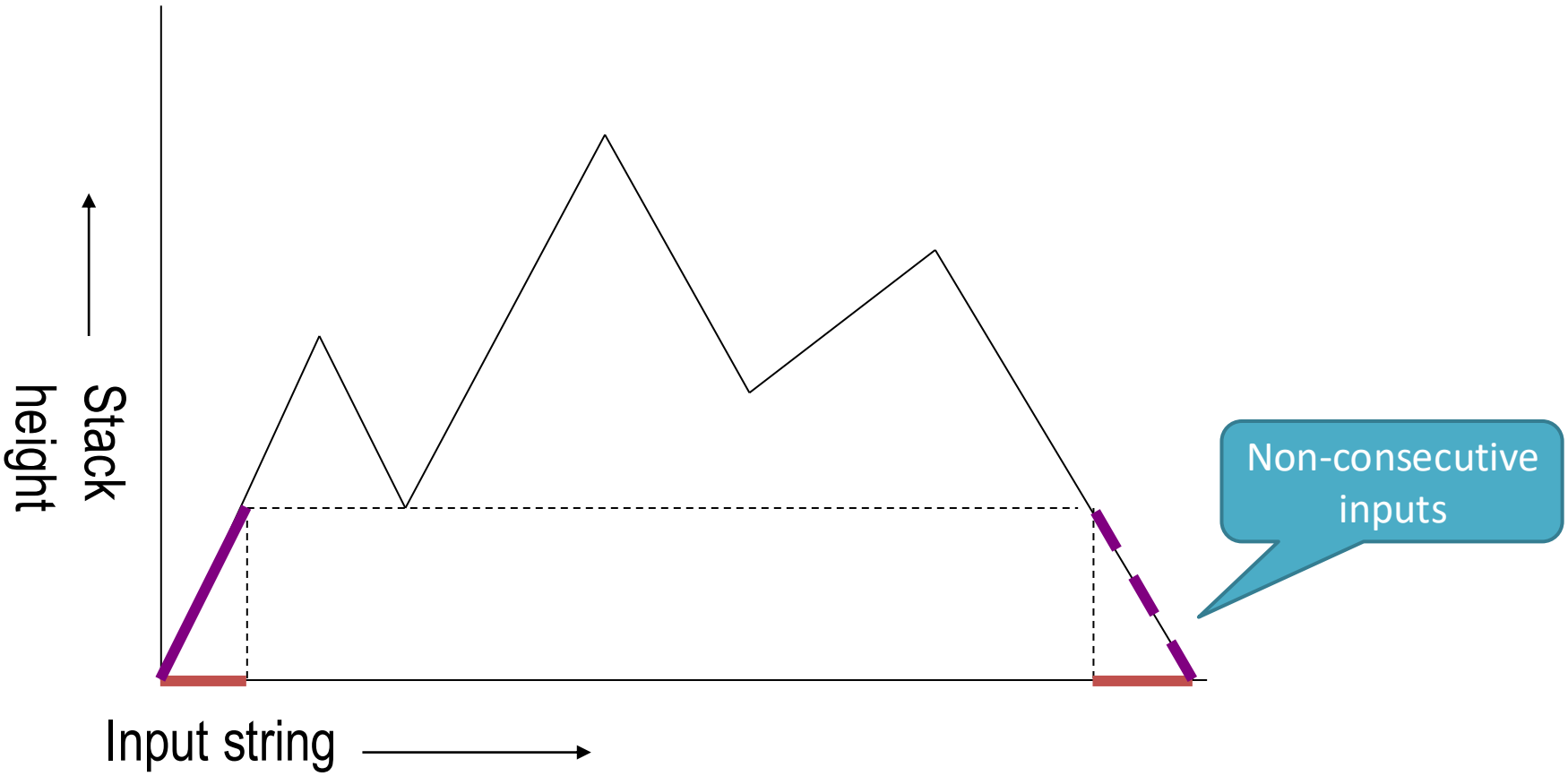
A language is CFL ⇐ some PDA recognizes it

- Case 1: the stack does not change (increase and then decrease) after some consecutive inputs

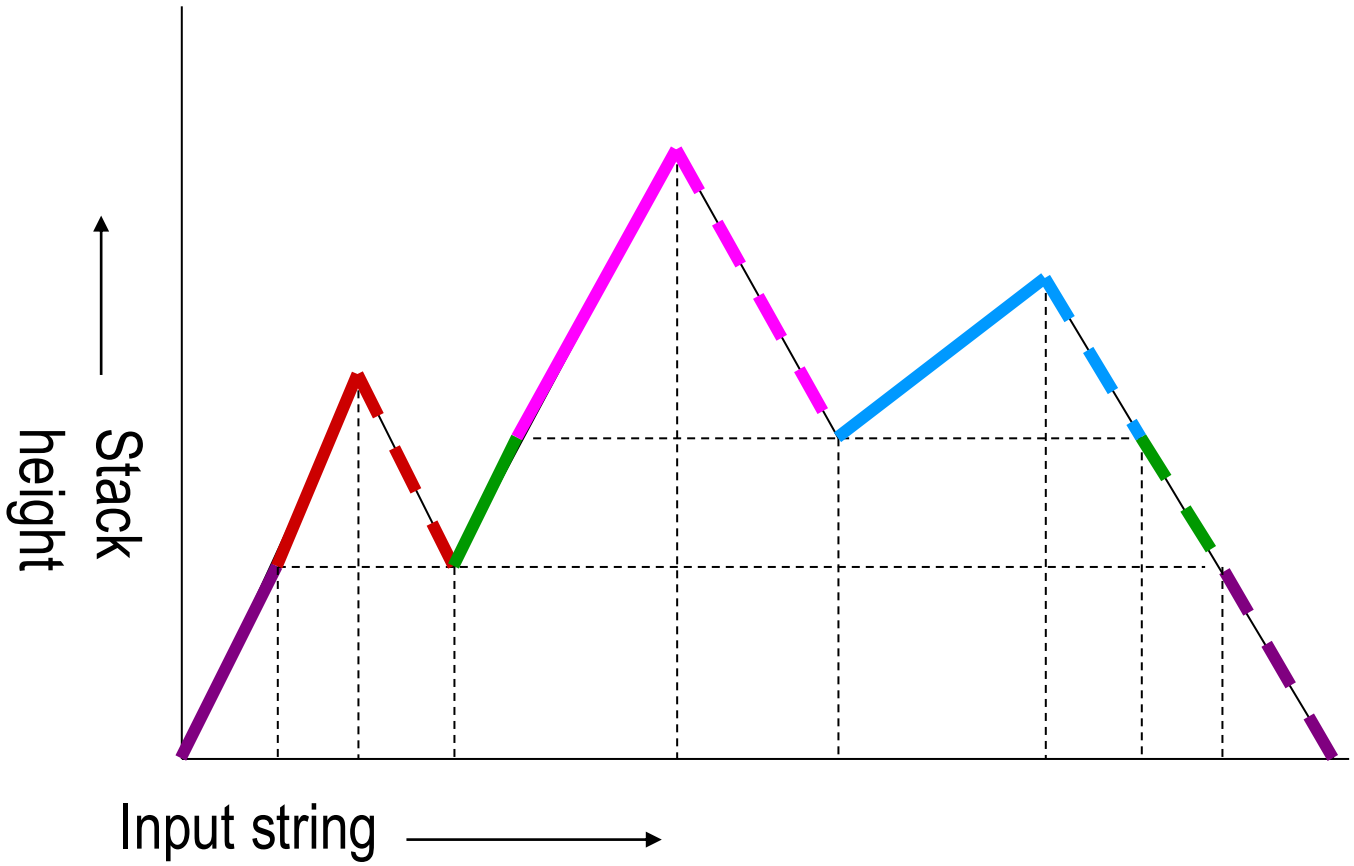


A language is CFL ⇐ some PDA recognizes it

- Case 2: the stack does not change (increase and then decrease) after some non-consecutive inputs



A language is CFL ⇐ some PDA recognizes it



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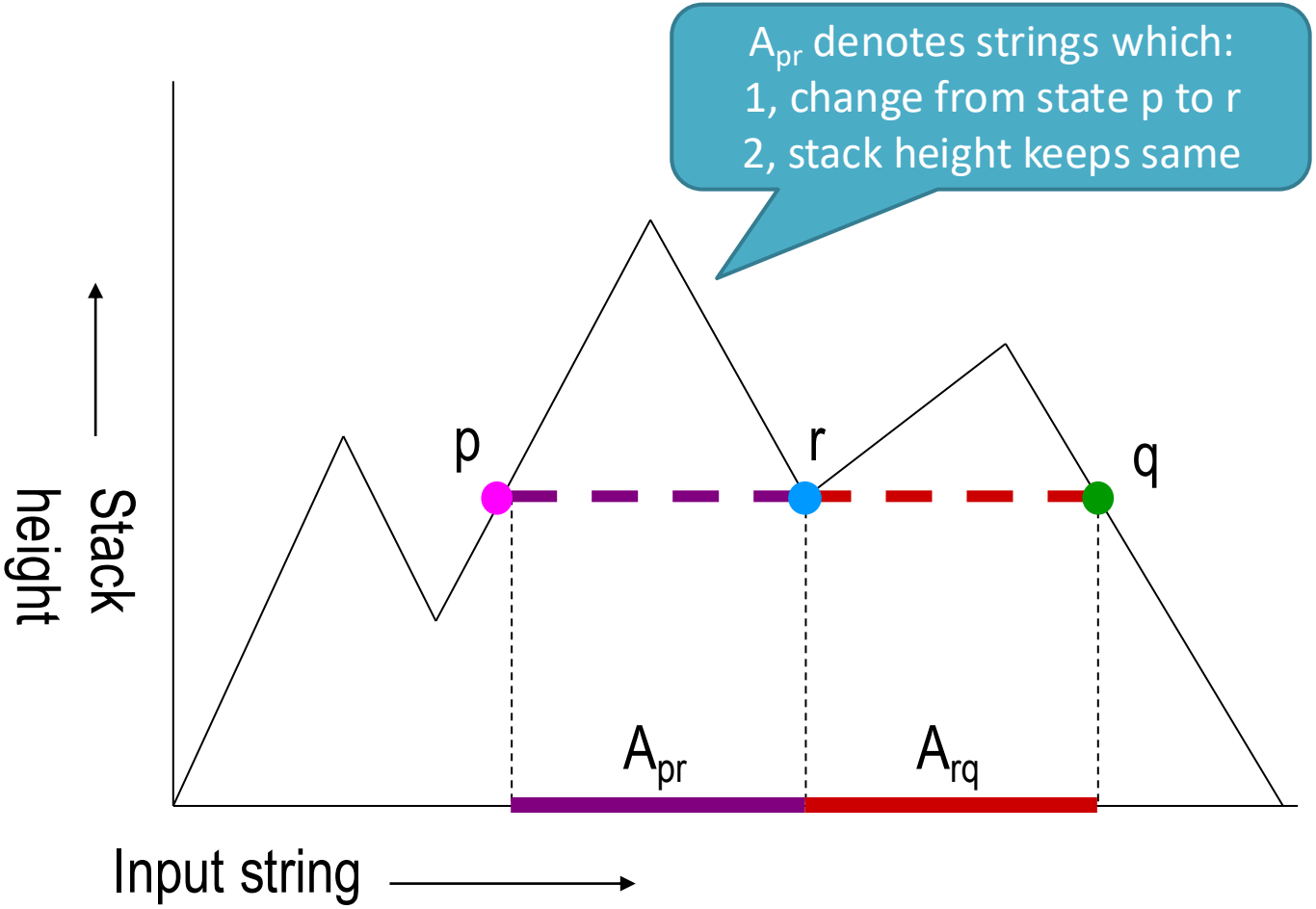
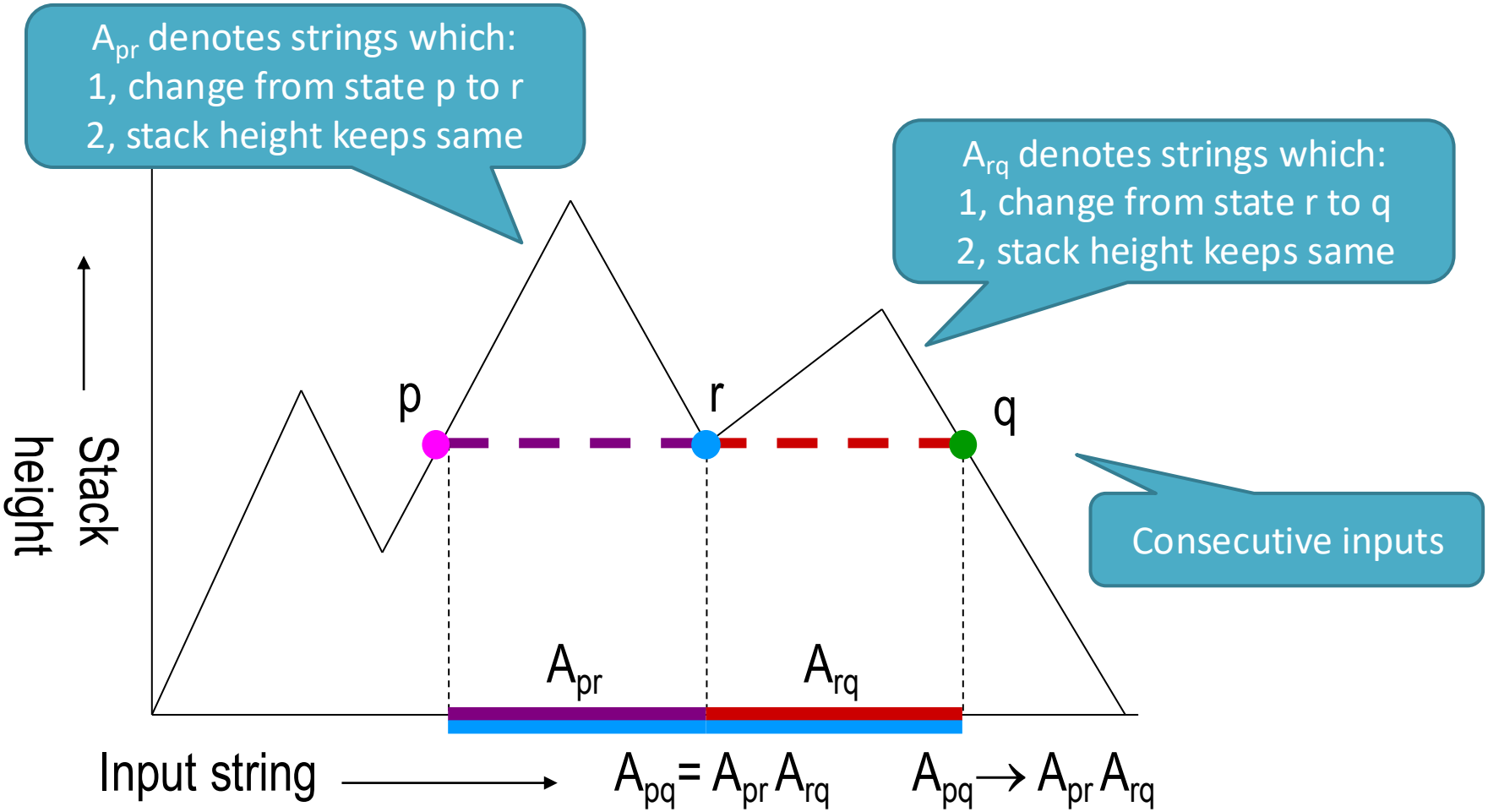


A language is CFL ⇐ some PDA recognizes it



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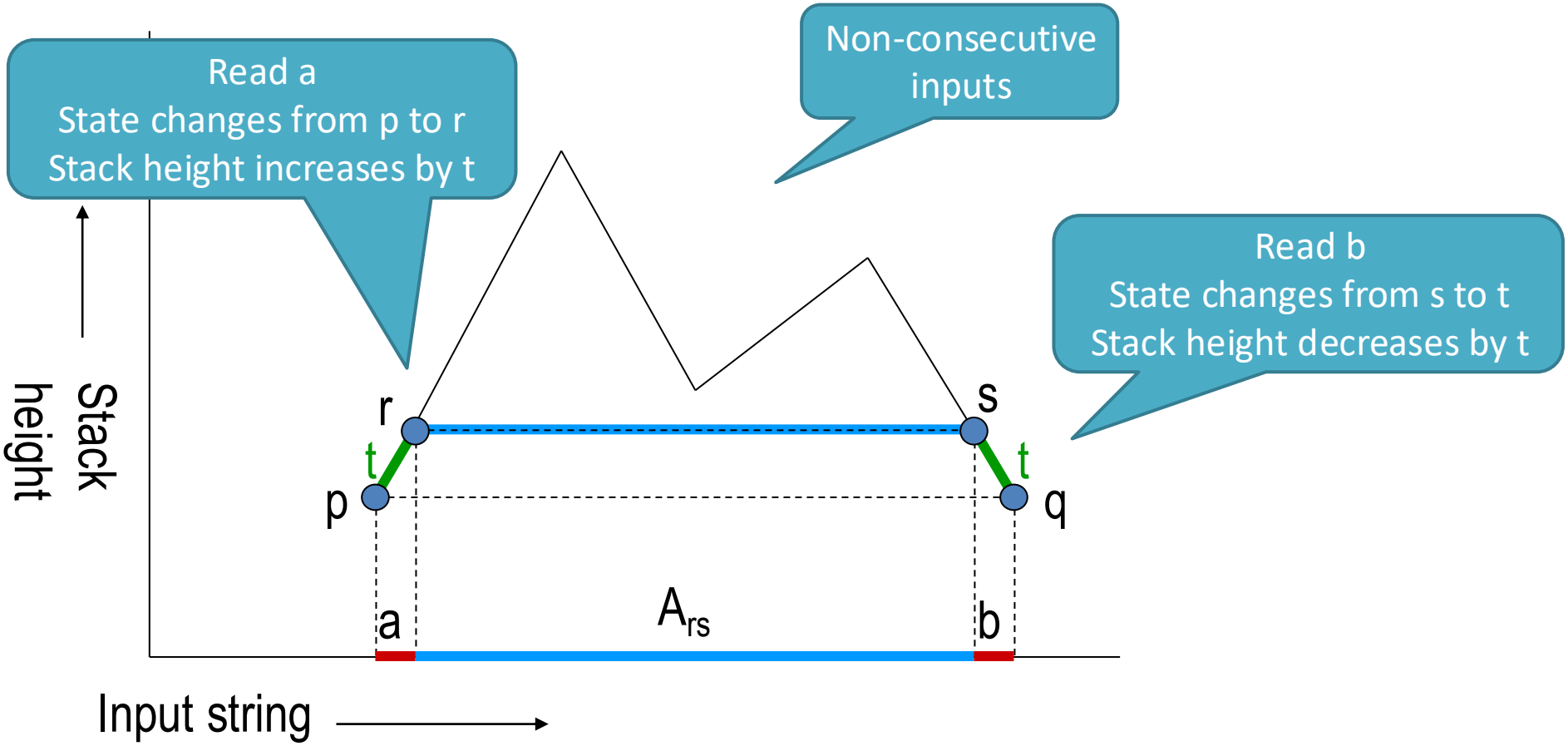
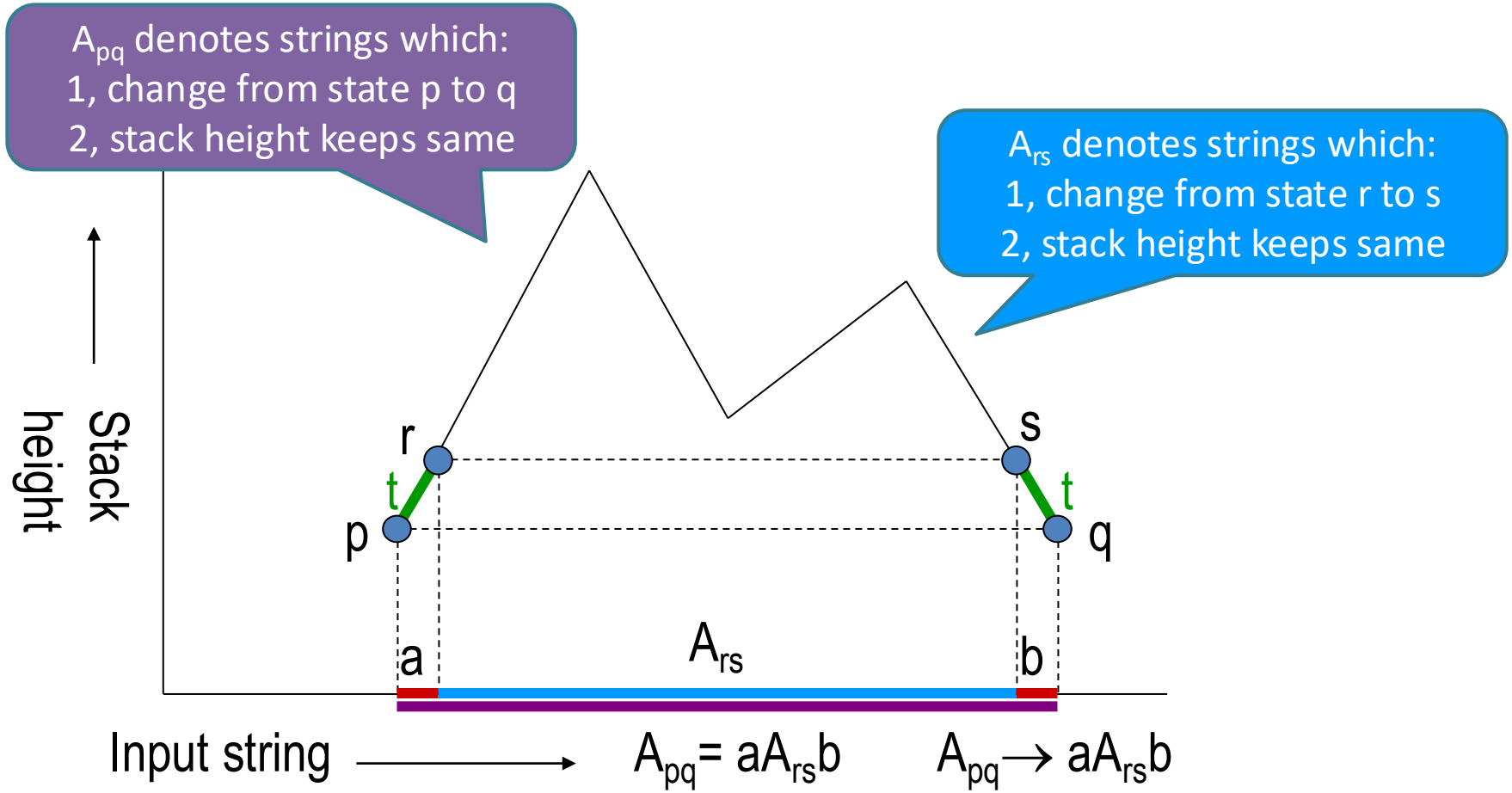


A language is CFL ⇐ some PDA recognizes it



CFL ⇐ PDA

A_{pq} denotes strings which:
1, change from state p to q
2, stack height keeps same

- For a PDA

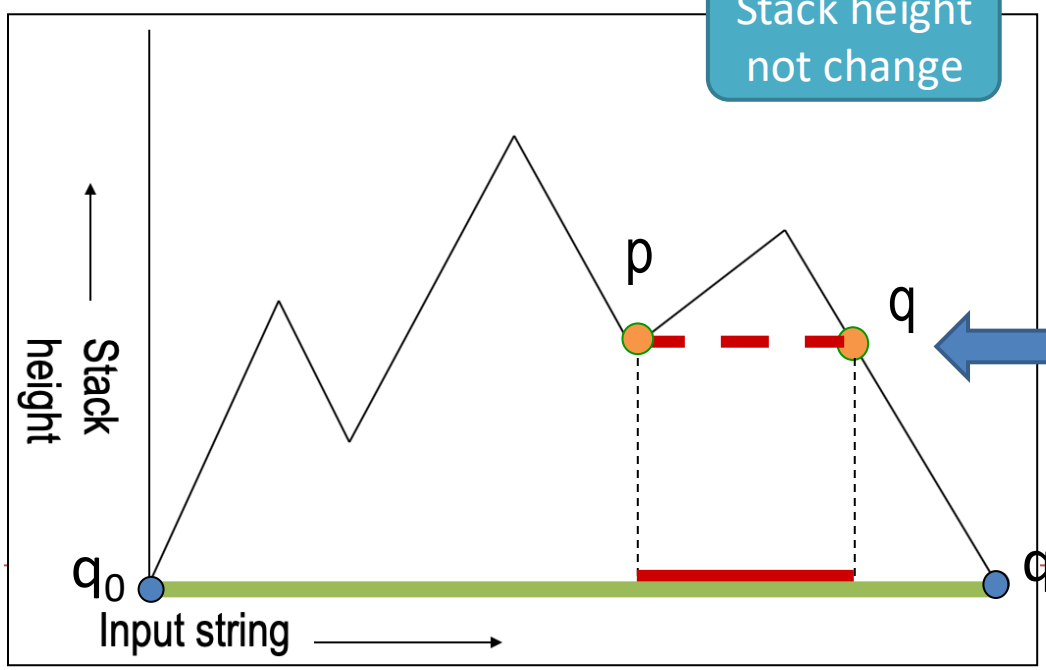
  $P=(Q, \Sigma, \Gamma, \delta, q_0, \{q_{accept}\})$,

  then create equivalent CFG

  $G=(\{A_{pq} \mid p,q \in Q\}, \Sigma, R, A_{q0qaccept})$,

Stack height not change

It defines where the PDA starts.



A_{pq}

$A_{q0qaccept}$

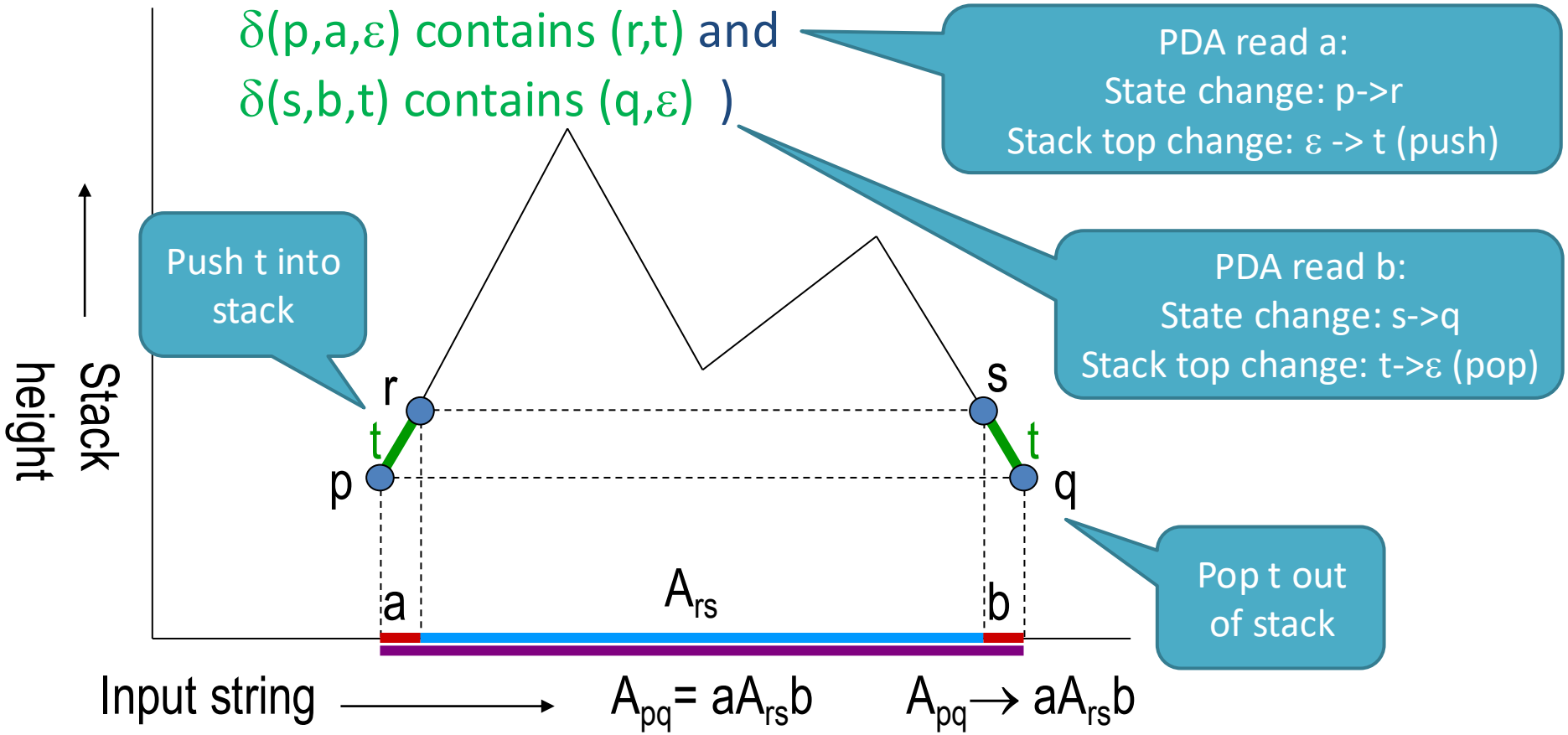
Non-Consecutive input rule

R has the following rules:

(1) $\mathbf{A_{pq} \rightarrow aA_{rs}b}$, (for each $p,q,r,s \in Q$, $a,b \in \Sigma_\varepsilon$, $t \in \Gamma$,

$\delta(p,a,\varepsilon)$ contains (r,t) and

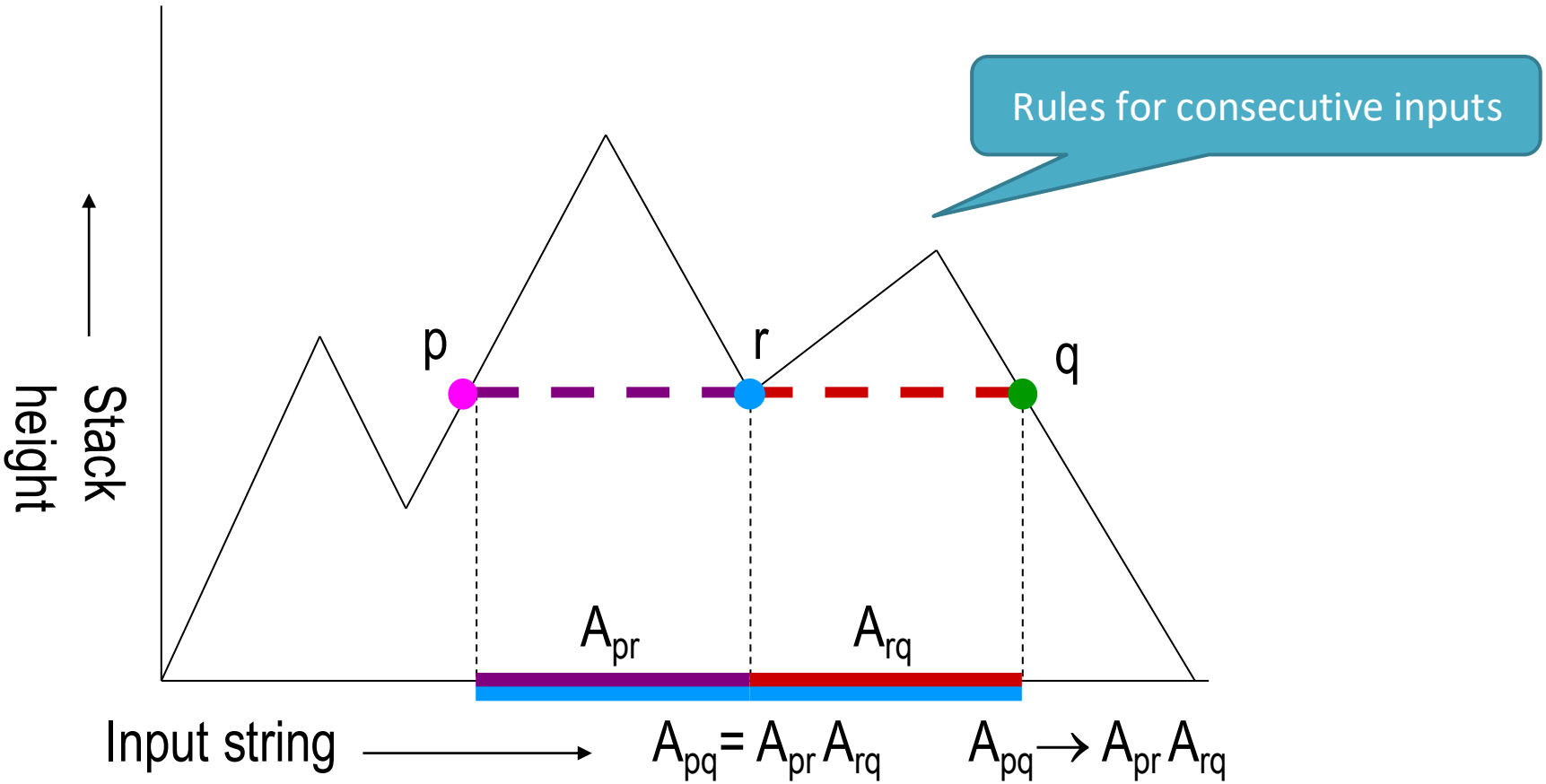
$\delta(s,b,t)$ contains (q,ε))



Consecutive input rule

R has the following rules:

(2) $A_{pq} \rightarrow A_{pr} A_{rq}$, (for each $p,q,r \in Q$)

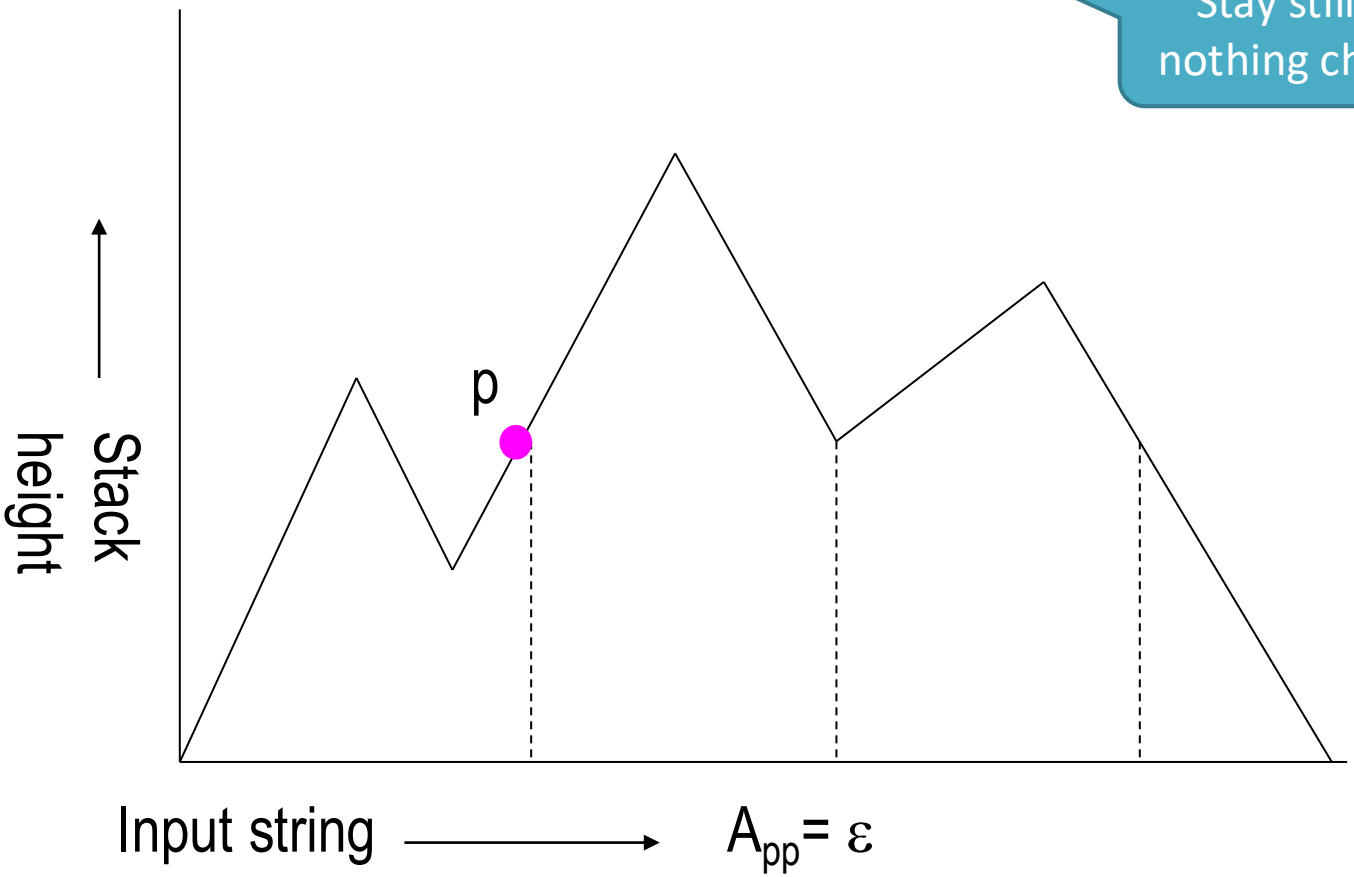


ε rule

R has the following rules:

(3) $A_{pp} \rightarrow \varepsilon$, (for each $p \in Q$)

Stay still and nothing changes



A language is CFL $\Leftarrow$ some PDA recognizes it

- For a PDA

  $P=(Q, \Sigma, \Gamma, \delta, q_0, \{q_{accept}\})$,

  then create equivalent CFG

  $G=(\{A_{pq} \mid p,q \in Q\}, \Sigma, R, A_{q0qaccept})$,

From start state $A_{q0qaccept}$, applying each rules the stack height will not change.

The strings accepted by PDA could be generated using grammar G

Conclusion

- Equivalence of PDA and CFG:
- A language is CFL $\Longrightarrow$ some PDA recognizes it
- A language is CFL $\Longleftarrow$ some PDA recognizes it